“SECURE-X
(Integrated with Helmet, Alcohol & Accident Detection)”

PROJECT REPORT

**SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENT FOR THE AWARD OF
DIPLOMA IN
ELECTRONICS & TELECOMMUNICATION ENGINEERING**

SUBMITTED BY

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**DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION
GOVERNMENT POLYTECHNIC, KARAD**

(2025 – 26)

GOVERNMENT POLYTECHNIC, KARAD

CERTIFICATE

This is to certify that,

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of final year Electronics & Telecommunication Engineering students have completed project of final year having title

"SECURE-X

(Integrated with Helmet, Alcohol & Accident Detection)"

during academic session 2025-2026 as a part of project work described by Government Polytechnic, Karad for partial fulfilment for the Diploma in Electronics & Telecommunication Engineering in the sixth semester.

The project work is the record of students own work under my guidance and to my satisfaction.

Dr. U. R. Deshmukh
Head of Department

Dr. S. A. Patil
Principal
Government Polytechnic, Karad

Mrs. U. S. Rangate
Guide

(External Examiner)

GOVERNMENT POLYTECHNIC, KARAD
CANDIDATE'S DECLARATION

I hereby certify that the work which is being presented in the project report entitled "SECURE-X (Integrated with Helmet, Alcohol & Accident Detection)" by us in partial fulfilment of requirements for the award of diploma in Electronics & Telecommunication Engineering submitted in the Department of Electronics & Telecommunication Engineering is an authentic record of our own work carried out during ODD/EVEN (2025-26) guided by Mrs. U. S. Rangate.

Signature and Name of the Students

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Mr. Meet Harkchand Shah (164580)

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Mr. Rupesh Sunil Patekar (164654)

This is to certify that the above statement made by the candidate is correct to the best of my/our knowledge.

Signature of the GUIDE

Mrs. U. S. Rangate

ACKNOWLEDGEMENT

I would like to place on record my deep sense of gratitude to Mrs. U. S. Rangate, Dept. of Electronics and Telecommunication for his generous guidance, help and useful suggestions.

I express my sincere gratitude to Dr. U. R. Deshmukh, head of Electronics and Telecommunication department, for his stimulating guidance, continuous encouragement and supervision throughout the course of present work.

I am extremely thankful to Dr. S. A. Patil, Principal, for providing me infrastructural facilities to work in, without which this work would not have been possible.

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1.Introduction

1.1. Problem Definition:-

Road safety plays a crucial role in protecting human life, and traffic authorities continuously implement various measures to reduce accidents. One of the major concerns worldwide is the increasing number of road accidents involving two-wheeler riders. Motorcyclists are highly vulnerable to severe injuries, especially head injuries, which often occur due to the absence of helmets or careless riding. Despite strict traffic regulations and awareness campaigns, many riders neglect safety measures, leading to fatal consequences. According to the World Health Organization (WHO), road traffic injuries are among the leading causes of death globally, particularly among young adults.

Helmets are designed to reduce the risk of head injuries; however, conventional helmets only provide passive protection and do not prevent accidents caused by factors such as alcohol consumption, over speeding, or delayed medical assistance after a crash. In many cases, accident victims do not receive immediate help because their location is unknown, resulting in loss of precious time during emergencies. Many lives could be saved if accidents were detected instantly and emergency contacts were notified without delay.

With the rapid advancement in embedded systems and IoT technology, smart safety solutions can be developed to enhance rider protection. To address these issues, we propose a Smart Helmet System that integrates advanced components such as an ESP32 microcontroller, alcohol sensor, GPS module, GSM module, accelerometer, and vibration sensor. This system ensures that the motorcycle starts only when the rider is wearing the helmet and is not under the influence of alcohol. In case of an accident, the helmet detects the impact and automatically sends the rider's location to emergency contacts.

1.2. Objectives of the study:-

The primary objective of the Smart Helmet System is to enhance the safety of two-wheeler riders by integrating modern embedded and IoT technologies into a conventional helmet. The system aims to ensure that the rider wears the helmet properly before the motorcycle engine can start, thereby promoting strict adherence to safety regulations. Another important objective is to prevent drunk driving by incorporating an alcohol sensor that detects the presence of alcohol in the rider's breath and restricts ignition if the level exceeds the permissible limit.
The project also focuses on automatic accident detection using sensors such as an accelerometer, gyroscope, and vibration sensor to identify sudden impacts or abnormal movements. In the event of a crash, the system is designed to immediately send an alert message along with the rider's real-time location through a GPS and GSM module to predefined emergency contacts. This helps in reducing response time and increasing the chances of survival by enabling faster medical assistance.
Additionally, the study aims to develop a cost-effective, reliable, and user-friendly safety device that can be easily implemented in real-world scenarios. By combining safety enforcement, accident detection, and emergency communication into a single integrated system, the Smart Helmet seeks to reduce road accident fatalities and contribute to safer transportation systems, as emphasized by organizations like the World Health Organization.

2. Literature Survey

- Smart Helmet with Accident Detection and Alert System:

Many researchers have proposed smart helmet systems that focus on accident detection using sensors like accelerometers and gyroscopes. These systems detect abnormal movements or impacts when a crash occurs and trigger an alert signal. The integration of GSM and GPS modules is commonly used to send the rider's location to emergency contacts. Studies show that such systems can significantly reduce emergency response time, potentially saving lives.

- Alcohol Detection and Prevention of Drunk Riding:

Alcohol-impaired driving is a major cause of road accidents globally. Research in this field has explored the use of MQ-series alcohol sensors integrated into helmets or vehicle systems to detect the presence of alcohol in breath. These studies highlight that early detection of alcohol can prevent ignition of the vehicle if alcohol levels exceed safe limits, thereby reducing accident risks associated with drunk riding.

- Helmet Wear Detection Techniques

Researchers have explored various methods for detecting whether a helmet is worn, including infrared (IR) sensors, pressure sensors, and magnetic switches. The goal is to confirm proper helmet usage before vehicle ignition. Findings indicate that these detection techniques can be effective in ensuring compliance with safety protocols and reducing injuries in case of accidents.

- Integration of Multiple Sensors for Accurate Detection

Several studies emphasize the advantage of using multiple sensors together (e.g., accelerometer, gyroscope, and vibration sensors) to improve accuracy in detecting crashes. Combining data from different sensors helps reduce false alarms and increases the reliability of the system in real-world conditions.

3. Current Scope

The current scope of the smart helmet project is rapidly expanding due to the use of modern technologies such as sensors, IoT, and wireless communication. Smart helmets are mainly used to improve safety, monitoring, and communication in different fields.

At present, the major scope of smart helmets is in **motorcycle safety systems**. Most smart helmets are designed to ensure helmet usage, prevent drunk driving, detect accidents, and send emergency alerts. Studies show that a large number of smart helmet applications focus on rider safety and accident rescue systems.

In addition, smart helmets are widely used in **health and safety monitoring**. They can measure parameters such as body temperature, fatigue, and other health conditions, helping to prevent accidents caused by human factors.

Another important area is **industrial safety**. Smart helmets are used in industries like construction and mining to detect harmful gases, monitor workers, and provide emergency alerts. Research shows that smart helmet applications are now focusing on both motorcyclists and workers for safety improvement.

The integration of **wireless communication technologies** like Bluetooth, Wi-Fi, and IoT allows real-time data transfer to mobile devices and cloud systems. Around 85% of smart helmets today use such communication technologies for monitoring and control.

Furthermore, smart helmets also provide **communication and convenience features** such as navigation, hands-free calling, and audio systems, improving user experience.

4. Methodology

4.1 Stepwise procedure of making project:-

1. Gather Components:
   - Acquire all necessary components, including ESP32, GSM800C, Relay module sensors, Arduino IDE, jumper wires, breadboard, power supply, and other supporting electronic components.
2. Hardware Setup:
   - The hardware setup begins with connecting the ESP32 as the main controller of the smart helmet system.
   - The IR sensor is connected to the ESP32 to detect whether the rider is wearing the helmet.
   - The MQ-3 Alcohol Sensor is connected to measure the alcohol level .
   - The MPU6050 sensor is connected using I2C communication (SDA and SCL pins) to detect tilt, motion, or fall of the helmet.
   - A vibration sensor is connected to detect strong shocks or impacts during an accident.
   - A push button is connected to allow the rider to manually send an emergency alert
   - All sensors are powered using the 3.3V or 5V supply from the ESP32 and connected with common GND.
   - After completing all connections, the ESP32 is connected to the computer using a USB cable to upload the program and test the system.
3. Sensor Calibration:
   - Sensor calibration is the process of adjusting a sensor so that it gives accurate and reliable readings.
   - During calibration, the sensor output is compared with a known standard value and adjustments are made.
   - Calibration helps to reduce measurement errors caused by environmental conditions or sensor drift.
   - For example, the MQ-3 Alcohol Sensor needs calibration to set the correct alcohol detection threshold.
   - Motion sensors like the MPU6050 are calibrated to remove offset errors in accelerometer and gyroscope readings.

- Calibration is usually done by recording sensor values in normal conditions and setting reference limits in the program.
- Proper calibration improves accuracy, reliability, and performance of the system.

4. ESP32 Programming:
    - The ESP32 is a microcontroller used to control sensors, devices, and IoT systems.
    - It is commonly programmed using the Arduino IDE.
    - A program mainly has two functions:
        i. setup() – runs once to initialize pins and sensors.
        ii. loop() – runs continuously to read sensors and control devices.
    - Pins are configured using pinMode() as INPUT or OUTPUT.
    - Sensor values are read using digitalRead() or analogRead().
    - Devices like LEDs, motors, or relays are controlled using digitalWrite().
    - Serial communication (Serial.begin) is used to monitor data on the serial monitor.
    - ESP32 supports communication protocols like I2C, UART, and SPI to connect sensors such as the MPU6050.

5. Testing and debugging:
    - Testing is the process of checking whether all components of the system are working correctly after the hardware setup and programming.
    - Each sensor such as the MQ-3 Alcohol Sensor, IR sensor, vibration sensor, and MPU6050 should be tested individually to verify their outputs.
    - The ESP32 is connected to the computer and the sensor values are observed using the Arduino IDE serial monitor.
    - If the system does not work properly, debugging is done to find and correct errors in the program or hardware connections.
    - Debugging may include checking wiring connections, correcting code errors, adjusting sensor thresholds, and verifying power supply.

6.User Interface Development:

- In a smart helmet system using the ESP32, the interface can be created using LEDs, buzzers, mobile notifications, or a display.
- Sensor data and system status can be monitored through the Arduino IDE serial monitor during testing.
- Alerts such as alcohol detection, helmet not worn, or accident detection can be shown to the user through visual or sound indicators.
- A mobile-based interface can also be developed to display data and alerts from the helmet system.

7.Integration and Optimization:

- Assemble all components into a compact and secure enclosure suitable for outer environments.
- Optimize the code for efficiency, ensuring low power consumption and fast data processing.
- Ensure reliable bluetooth connectivity to avoid data loss or delays in alerts.

8.Documentation:

- Document the hardware setup, circuit diagrams, and ESP32 code for future reference.
- Prepare a user manual detailing system setup, usage instructions, and troubleshooting steps.

9.Field Testing:

- Field testing is the process of testing the system in a real environment to check its performance.
- In the smart helmet project using the ESP32, the device is tested while riding or simulating real riding conditions.
- Sensors such as the MQ-3 Alcohol Sensor, IR sensor, vibration sensor, and MPU6050 are checked to see if they respond correctly.
- Accident detection, helmet detection, and alcohol detection functions are verified in practical conditions.
- The system performance, accuracy, and response time are observed during testing.
- Any problems found during field testing are corrected to improve the reliability and efficiency of the system.

10.Deployment and Maintenance:

- Development is the process of designing and building the system using the ESP32 and sensors.
- The program is written and uploaded using the Arduino IDE.
- Maintenance involves checking the system regularly and fixing hardware or software issues.
- It ensures the system works properly and reliably for a long time.

4.2 Details of software tools used:-

1. Arduino IDE:-
   Arduino IDE (Integrated Development Environment) is an open-source software used to write, compile, and upload programs to microcontrollers like ESP32 or Arduino boards.

- In the Smart Helmet project, Arduino IDE is used to:
  a. Program the ESP32 microcontroller
  b. Interface sensors like MQ-3 alcohol sensor, MPU6050, vibration sensor
  c. Control GSM and GPS modules

2. Proteus:-
   Proteus is a simulation software used for designing and testing electronic circuits before hardware implementation.

- In this project, Proteus is used to:
  a. Design the circuit diagram
  b. Simulate microcontroller-based circuits
  c. Test sensor and module connections virtually
  d. Reduce hardware errors

3. Visual Studio Code (VS Code):-
   Visual Studio Code is a lightweight but powerful source-code editor developed by Microsoft.

- In this project, VS Code is used to:
  a. Write and manage embedded system code
  b. Use PlatformIO extension for ESP32 programming
  c. Organize project files efficiently
  d. Debug code more professionally

4. KiCad:-
   KiCad is an open-source Electronic Design Automation (EDA) tool used for schematic design and PCB layout.

- In the Smart Helmet project, KiCad is used to:
  a. Create schematic diagrams
  b. Design PCB layout for compact helmet circuitry
  c. Generate Gerber files for PCB manufacturing

4.3 Block diagram:-

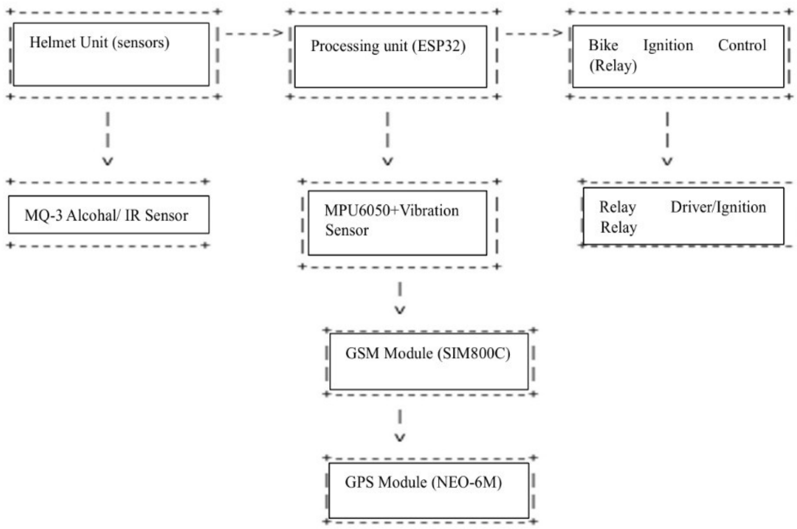


Fig.4.3 Block diagram

- Helmet Unit (Sensors):
  - The helmet contains sensors such as MQ-3 alcohol sensor and IR sensor. The IR sensor checks whether the rider is wearing the helmet, and the MQ-3 sensor detects alcohol from the rider's breath.
- Processing Unit (ESP32):
  - All sensor data is sent to the ESP32 microcontroller, which processes the inputs and decides whether the bike should start or stop.
- MPU6050 and Vibration Sensor:
  - These sensors detect sudden tilt, fall, or accident conditions by measuring motion and vibration.
- Ignition Control (Relay):
  - The ESP32 controls a relay connected to the bike's ignition system.
  - If the helmet is worn and no alcohol is detected → ignition is enabled.
  - If alcohol is detected or helmet is not worn → ignition remains OFF.
- GSM Module (SIM800C):
  - In case of an accident, the GSM module sends an SMS alert to emergency contacts.
- GPS Module (NEO-6M):
  - The GPS module provides the exact location of the rider, which is sent in the alert message.

4.1.1 Esp 32:-

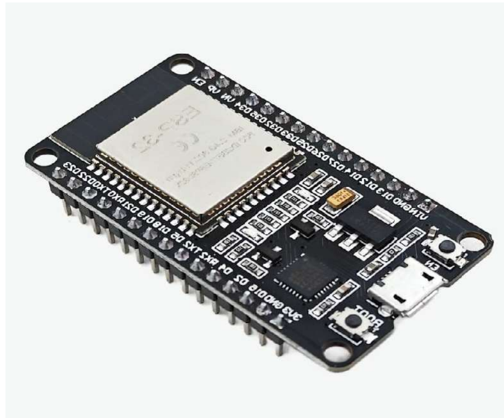


Fig.4.1.1 Esp32

The ESP32 is a powerful, low-cost microcontroller developed by Espressif Systems. It is widely used in IoT, embedded systems, automation, and wireless communication projects.

- Specification:

1. Developed by Espressif Systems
2. 32-bit Dual-Core Microcontroller:
    a. Xtensa LX6 processor
    b. Up to 240 MHz clock speed
3. Built-in Wi-Fi:
    a. 802.11 b/g/n
    b. Used for IoT applications
4. Built-in Bluetooth:
    a. Bluetooth Classic
    b. BLE (Low Energy)
5. Large Number of GPIO Pins:
    a. 30+ programmable pins
    b. Supports Digital I/O
6. Supports Multiple Communication Protocols:
    a. UART
    b. SPI
    c. I2C
    d. PWM
    e. ADC-DAC

7. Memory:
    a. 520 KB SRAM
    b. 4MB external Flash (typical module)
8. Operating Voltage:
    a. 3.3V logic
    b. 5V input via VIN/USB
9. Low Power Modes:
    a. Active mode
    b. Light sleep
    c. Deep sleep
10. High Processing Speed:
    a. Much faster than Arduino UNO

4.1.2 GSM 800C:-

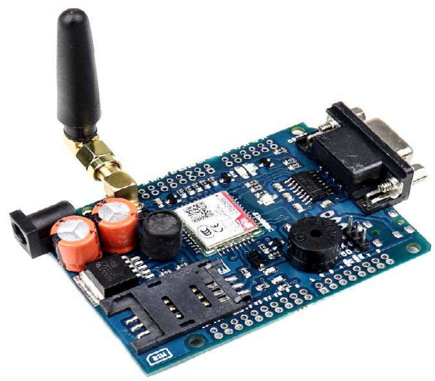


Fig.4.1.2 GSM 800C

SIM800C is a quad-band GSM/GPRS module used for SMS, voice call, and internet communication. It is controlled using AT commands and operates on a 2G & 3G network.

- Specification:

1. It is a GSM/GPRS communication module
2. Used for:
   a. Sending SMS
   b. Making voice calls
   c. Receiving calls
3. Operates on 2G GSM network
4. Quad Band Support:
   a. 850 MHz
   b. 900 MHz
   c. 1800 MHz
   d. 1900 MHz

5. Controlled using AT Commands
6. Communication with microcontroller via:
   a. UART (TX/RX)
7. Requires SIM card to operate

8. Operating Voltage:
    a. 3.8V – 4.2V (Recommended 4.0V)
9. High Current Requirement:
    a. Idle: ~20mA

During Call/SMS burst: up to 2A

10. Supports:
    a. SMS (Text & PDU mode)
    b. Voice call
    c. GPRS data
    d. Caller ID

4.1.3 GPS NEO6M:-

Fig.4.1.3. GPS NEO6M

GPS NEO-6M is a small and popular GPS receiver module used in electronics projects to get location coordinates (latitude, longitude), speed, and time from GPS satellites.
The NEO-6M module receives signals from **GPS satellites** and calculates the device's position on Earth.

- Specifications:

1. GPS Chipsetu-blox NEO-6M
2. Frequency Band:
    a. L1 – 1575.42 MHz
3. Position Accuracy ~2.5 m
4. Velocity Accuracy 0.1 m/s
5. Update Rate:
    a. 1 Hz (default), up to 5 Hz
6. Cold Start Time ~27 s
7. Warm Start Time ~25 s
8. Hot Start Time ~1 s
9. Tracking Sensitivity:
    a. −161 dBm
10. Operating Voltage:
    a. 3.3 V – 5 V
11. Communication InterfaceUART (TX, RX)

12. Default Rate:
    a. 9600 bps(Default)
    b. 230400 bps(Maximum)

13. Protocols Supported:
    a. NMEA
    b. UBX
14. Operating Temperature:
    a. −40°C to +85°C
15. Antenna:
    a. 25×25 mm ceramic GPS antenna
16. Backup Battery:
    a. CR1220 for RTC and faster startup

4.1.4 Sensors:-

1. IR Sensor:

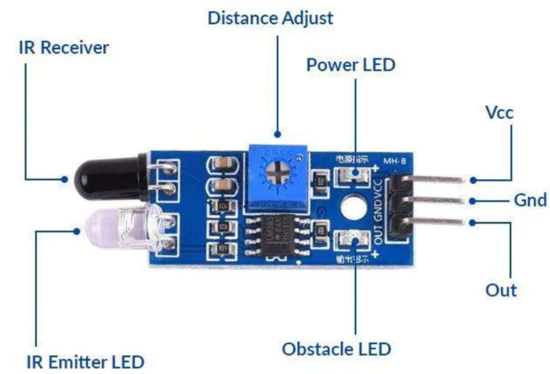


Fig.4.1.4.1. IR Sensor

An IR Sensor (Infrared Sensor) is an electronic sensing device that detects infrared radiation in the surrounding environment and converts it into an electrical signal. Infrared radiation is a type of electromagnetic wave with a wavelength longer than visible light (about 700 nm to 1 mm), which cannot be seen by the human eye

- Specification:

a. Operating Voltage: Usually 3.3 V – 5 V DC
b. Detection Range: About 2 cm to 30 cm (depends on sensor type)
c. Operating Current: Around 10 – 30 mA
d. Output Type: Digital or Analog output
e. Infrared Wavelength: Typically 850 nm – 940 nm
f. Response Time: Very fast (in milliseconds)
g. Detection Angle: Around 30° – 60°

2. MQ3 alcohol sensor:

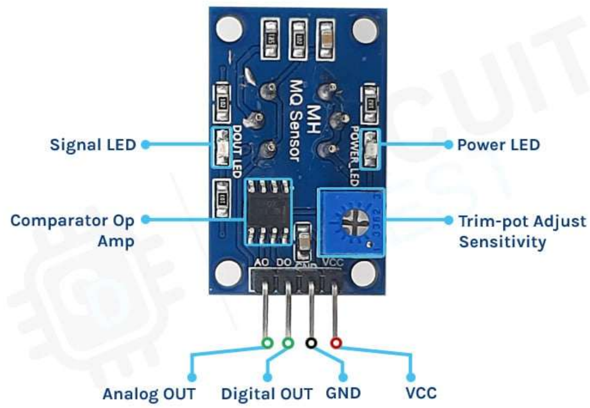


Fig.4.1.4.2. MQ3 alcohol sensor

The MQ-3 alcohol sensor is a gas sensor used to detect alcohol vapors (ethanol) in the air. It is commonly used in breath analyzer devices to measure whether a person has consumed alcohol.

- Specification:

a. Operating Voltage: 5 V
b. Heater Voltage: 5 V
c. Heater Current: ~150 mA
d. Power Consumption: About 750 mW
e. Detection Range: 0.04 mg/L – 4 mg/L alcohol in air
f. Output Type: Analog and Digital
g. Preheat Time: Around 20 seconds
h. Operating Temperature: −10°C to 50°C
i. Sensitive Material: Tin dioxide (SnO_2)

3. MPU6050 sensor:

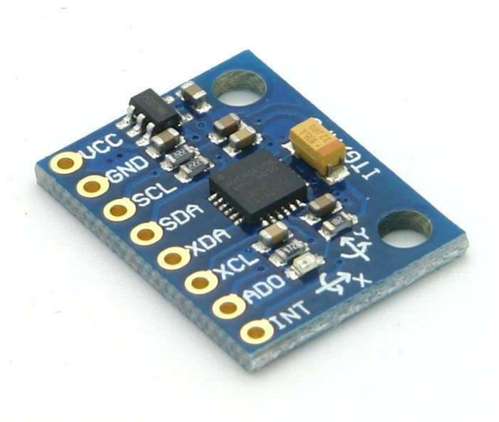


Fig.4.1.4.3. MPU6050 sensor

The MPU6050 is a 6-axis motion tracking sensor used to measure movement, tilt, and rotation of an object. It combines a 3-axis accelerometer and a 3-axis gyroscope in a single chip.

- Specification:

a. Operating Voltage: 3V – 5V
b. Communication Protocol: I2C
c. Axes: 3-axis accelerometer + 3-axis gyroscope
d. Accelerometer Range: ±2g, ±4g, ±8g, ±16g
e. Gyroscope Range: ±250, ±500, ±1000, ±2000 °/s
f. Digital Output: 16-bit ADC
g. Operating Temperature: −40°C to 85°C

4. Shock sensor:

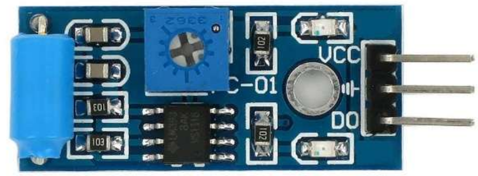


Fig.4.1.4.4. Shock sensor

A Shock Sensor is an electronic sensor that detects sudden vibration, impact, or shock. When a physical shock occurs (like hitting, tapping, or vibration), the sensor generates a signal that a microcontroller such as Arduino can read.

- Specification:

a. Operating Voltage: 3.3 V – 5 V DC
b. Operating Current: ≈ 15 mA
c. Output Type: Digital (HIGH / LOW)
d. Comparator IC: LM393
e. Sensor Type: Vibration / Shock detection sensor
f. Detection Method: Mechanical vibration switch
g. Trigger Level: Adjustable threshold
h. Response Time: Few milliseconds (very fast)

4.1.5 Converter:-

1. DC-DC Boost Converter (MT3608 Step-Up Converter):

Fig.4.1.5.1. Boost converter

A DC-DC Boost Converter MT3608 is a small electronic module used to increase (step-up) a low DC voltage to a higher DC voltage.

- Specifications:

a. Input Voltage: 2V – 24V DC
b. Output Voltage: 5V – 28V DC (adjustable)
c. Maximum Output Current: up to 2A (recommended ≈1A for stable operation)
d. Conversion Efficiency: up to 93%
e. Switching Frequency: 1.2 MHz

2. DC-DC Buck Converter (LM2596 Step-Down Converter):

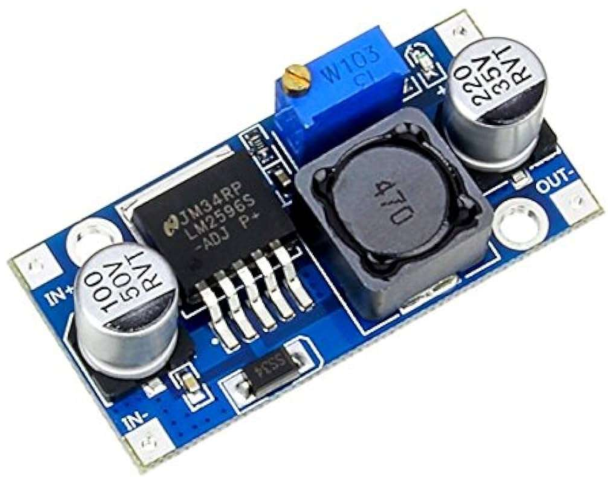


Fig.4.1.5.2. Buck converter

A DC-DC Buck Converter LM2596 is an electronic module used to reduce (step-down) a higher DC voltage to a lower DC voltage while maintaining good efficiency.

- Specification:

a. Input Voltage: 4 V – 40 V DC
b. Output Voltage: 1.25 V – 37 V DC (adjustable)
c. Maximum Output Current: up to 3 A
d. Efficiency: up to 92%
e. Switching Frequency: 150 kHz

4.1.6 Realay module:-

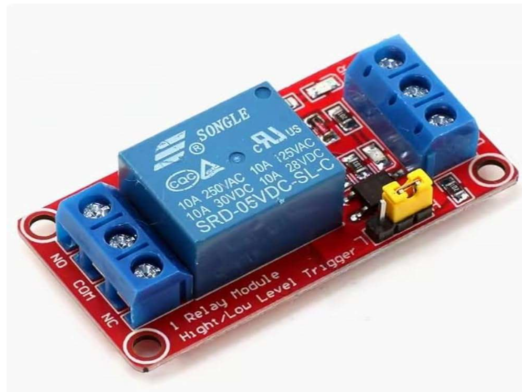


Fig.4.1.6.1 Relay module

A Relay Module is an electronic switching device used to control high-voltage or high-current devices using a low-voltage signal. It acts like an electrically operated switch.

- Specification:

a. Operating Voltage: 5 V DC
b. Trigger Voltage: 3.3 V – 5 V
c. Rated Load Voltage:
   i. 250 V AC
   ii. 30 V DC
d. Maximum Load Current: 10 A
e. Coil Power Consumption: ≈ 0.36 W

4.1.7 Other components:-

1. Push button:

Fig.4.1.7.1 Push button

A 4×4 mm Push Button Switch (also called a tactile switch) is a small momentary switch used in electronic circuits to open or close a circuit when pressed.

2. Resistor:

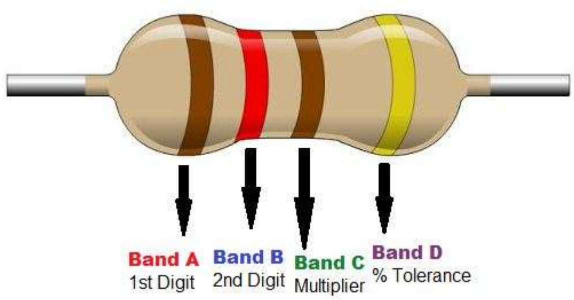


Fig.4.1.7.2 Resistor

A resistor is a basic electronic component used to limit or control the flow of electric current in a circuit. It provides resistance, which reduces the current passing through a circuit.

3. Capacitor:

Fig.4.1.7.3 Capacitor

A capacitor is a passive electronic component used to store electrical energy in the form of an electric field. It releases this stored energy when required in a circuit.

4. PCB:

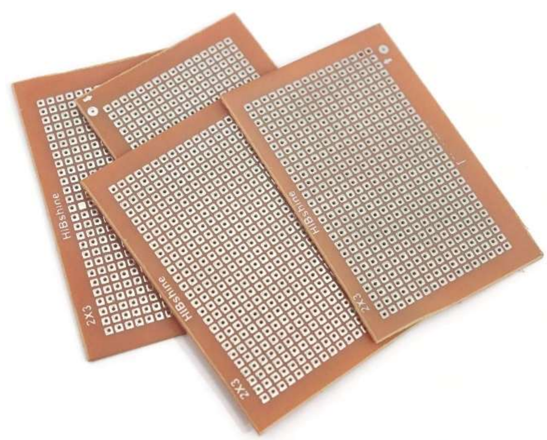


Fig.4.1.7.4 PCB

A PCB (Printed Circuit Board) is a board used to mechanically support and electrically connect electronic components using conductive copper tracks, pads, and traces. It replaces messy wire connections and makes circuits compact, reliable, and organized.

5. 18650 lithium ion battery:

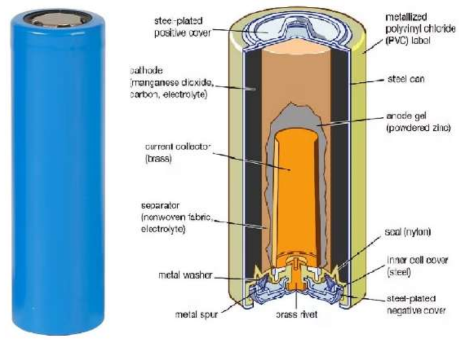


Fig.4.1.7.5. Battery

An 18650 Lithium-Ion Battery is a rechargeable cylindrical battery commonly used in power banks, laptops, flashlights, electric vehicles, and electronic projects.

- Specification:

a. Nominal Voltage: 3.7 V
b. Maximum Voltage (Fully Charged): 4.2 V
c. Cut-off Voltage (Discharged): about 2.5 V – 3.0 V
d. Capacity: typically 2000 mAh – 2500 mAh
e. Maximum Discharge Current: 5 A – 20 A (depends on battery type)

6. Battery charger module TP4056:

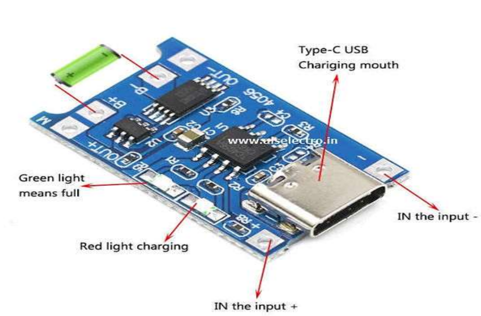


Fig.4.1.7.6. Battery charger module

A TP4056 Battery Charger Module is a small charging circuit used to safely charge a single 3.7V lithium-ion battery (like an 18650 battery). It uses a constant current / constant voltage (CC/CV) charging method to charge the battery safely.

7. Jumper wires:

Fig.4.1.7.7 Jumper wires

Jumper wires are small electrical wires used to connect components together on a breadboard, PCB, or between electronic modules without soldering.

Type	Connection
Male to Male (M-M)	Breadboard to breadboard
Male to Female (M-F)	Module to breadboard
Female to Female (F-F)	Module to module

8. Header pins:

Fig.4.1.7.8 Header pins

Header pins are metal connector pins mounted on a plastic strip used to connect electronic components, PCBs, or modules together.

4.1.8 Circuit diagram:-

1. Helmet module:

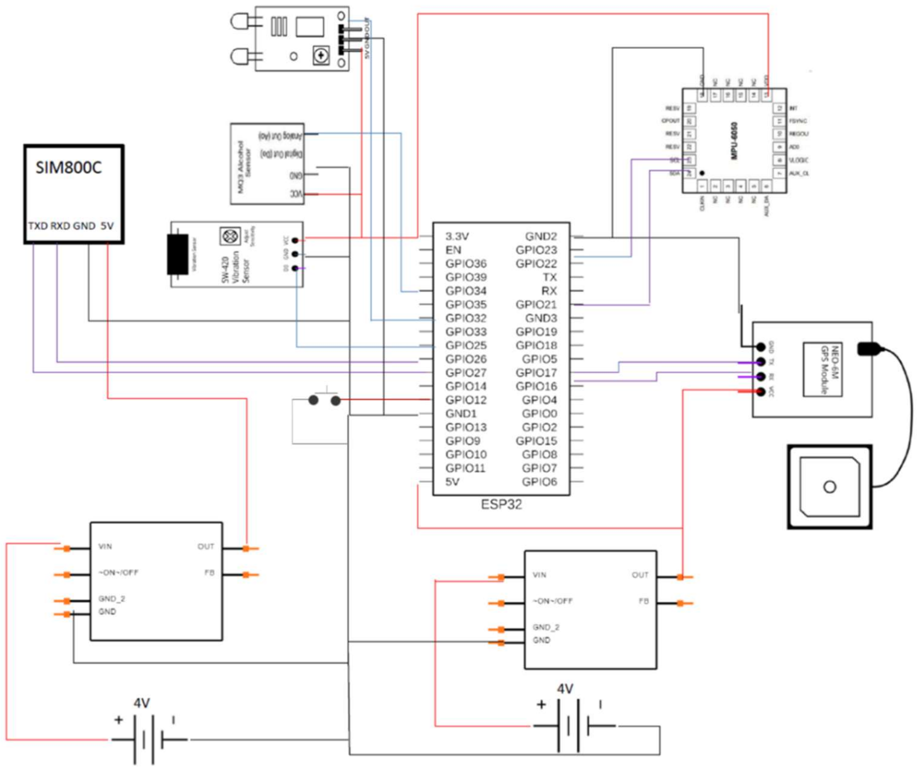


Fig.4.1.8.1 Circuit diagram(Bike module)

1. Power Supply:
   - The smart helmet system requires a stable and continuous power source for reliable operation. The ESP32 is powered through its VIN pin using a regulated power supply, while the GND is connected to the common ground. A backup battery is connected in parallel to ensure uninterrupted functioning during power failure. This dual power setup guarantees continuous monitoring and safety features, making the system dependable for real-time applications.
2. Sensors Integration (Alcohol, IR, Shock, MPU6050):
   - The smart helmet integrates multiple sensors to enhance rider safety:
   - Alcohol Sensor (MQ-3/MQ-135): Detects alcohol levels from the rider's breath. The analog output is connected to an ESP32 ADC pin. If the value exceeds a predefined threshold, the system prevents ignition or sends an alert.

- IR Sensor: Used to detect whether the helmet is worn properly. It is connected to a digital GPIO pin of the ESP32. The system ensures the vehicle starts only when the helmet is worn.
- Shock/Vibration Sensor: Detects accidents or sudden impacts. It is connected to a digital input pin. When abnormal vibration is detected, it triggers an emergency response.
- MPU6050 (Accelerometer & Gyroscope): Connected via I2C communication (SDA, SCL) to the ESP32. It monitors tilt, orientation, and sudden movement to detect accidents accurately.

3. GPS and GSM Communication (NEO-6M & SIM800C):
    - The system uses communication modules for real-time tracking and alerts:

- GPS Module (NEO-6M): Provides real-time location coordinates (latitude and longitude). It communicates with ESP32 using UART (TX/RX pins).
- GSM Module (SIM800C): Sends SMS alerts during emergencies. It is connected to ESP32 via serial communication. When an accident is detected, it sends the rider's location to predefined emergency contacts.

4. Working Process (Step-by-Step):
    1. The ESP32 continuously reads data from all sensors (alcohol, IR, shock, MPU6050).
    2. It checks safety conditions:
        - Helmet is worn
        - No alcohol detected
    3. If not detected bike get start
    4. In case of an accident (shock + abnormal tilt detected):
        - GPS fetches location
        - GSM sends emergency SMS
5. Final Overview:
    - This smart helmet system enhances rider safety using multiple sensors, real-time monitoring, and communication technologies. The integration of alcohol detection, helmet detection, accident detection, and GPS-GSM alert system ensures quick emergency response. The addition of Bluetooth connectivity and buzzer alerts further improves usability. With a reliable dual power supply, the system operates continuously, making it an efficient and practical solution for road safety applications.

2. Bike module:

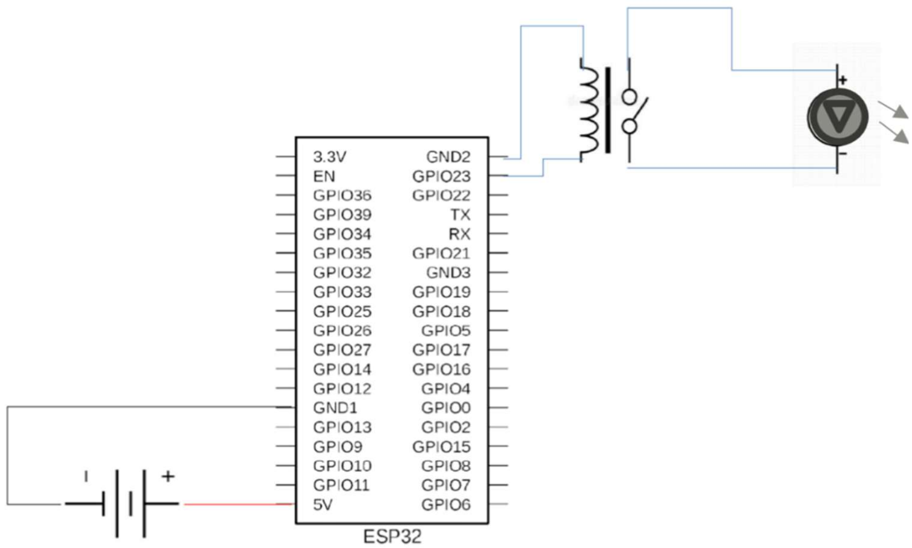


Fig.4.1.8.2 Circuit diagram(Bike module)

1. Power Supply
   - The ESP32 is powered using a DC battery supply connected to its 5V (VIN) and GND pins.
   - This powers the microcontroller and enables it to control external devices.
2. Relay Control by ESP32
   - A relay is connected to one of the GPIO pins of the ESP32.
   - The relay acts as an electrical switch controlled by the ESP32.
   - When the ESP32 sends a HIGH signal to the relay input, the relay gets activated (energized).
   - When the signal is LOW, the relay remains OFF.
3. Relay Switching Mechanism
   - The relay has two parts:
     - Coil (control side) connected to ESP32
     - Switch contacts (load side) connected to motor circuit
   - When the coil is energized:
     - The internal switch moves from Normally Open (NO) to Closed
     - This completes the circuit for the motor

4. Motor Operation

- The motor is connected to an external power supply through the relay.
- When the relay is ON:
  - Circuit is completed
  - Motor receives power and starts rotating
- When the relay is OFF:
  - Circuit is open
  - Motor stops

4.2 Working model:-

1. Initialization:
    - The system starts by powering up the ESP32, sensors (alcohol, IR, shock, MPU6050), GPS module, GSM module, and other connected components.
    - The ESP32 initializes all sensors and communication modules (Bluetooth, GPS, GSM).
    - Serial communication is established for GPS and GSM modules to ensure proper data transmission.
    - The MPU6050 sensor is calibrated to accurately detect tilt, motion, and orientation.

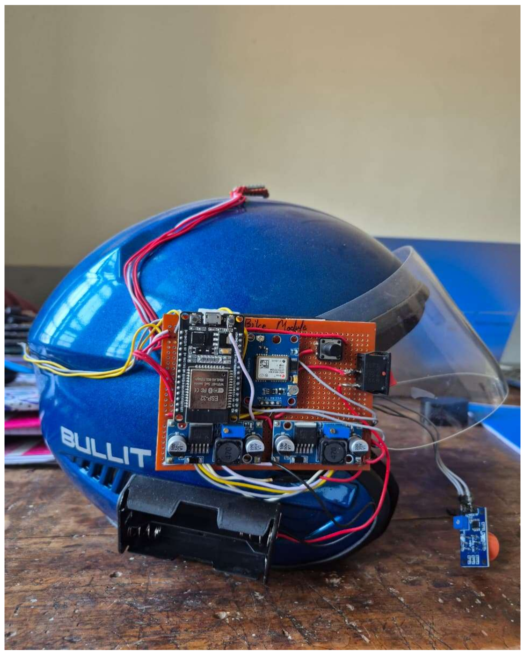
2. Safety Condition Monitoring:
    - The system continuously monitors whether the helmet is worn using the IR sensor.
    - The alcohol sensor checks the rider’s breath for alcohol content.
    - The MPU6050 tracks head movement, tilt, and abnormal angles.
    - The shock sensor detects sudden impacts or vibrations indicating a possible accident.
    - The ESP32 processes all sensor data and checks if conditions are safe for riding.

3. Alert and Prevention System:
    - If the helmet is not worn, the system prevents vehicle ignition or triggers a warning.
    - Alerts continue until the unsafe condition is resolved.

4. Emergency Detection and Response:
    - In case of an accident (detected via shock sensor and abnormal tilt from MPU6050), the system triggers an emergency response.
    - The GPS module fetches real-time location coordinates.
    - The GSM module sends an SMS alert with location details to predefined emergency contacts.
    - This ensures quick medical assistance and reduces response time.

5. Response and Action:

- When an alert is received (via SMS or mobile device), emergency contacts can take immediate action.
- The system reduces response time in accidents and improves rider safety.
- Continuous monitoring ensures prevention, detection, and quick response to dangerous situations.



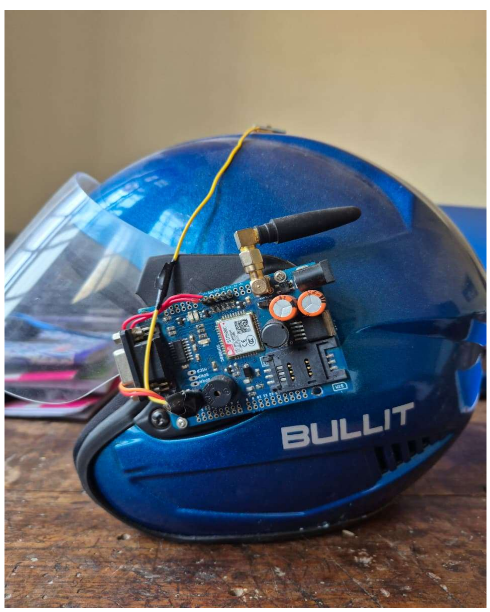


Fig. Working Model

4.3 Battery backup calculation:

1. Power Management Calculation – Helmet Unit

- Battery capacity:2500 mAh
  a. Number of cells:2
  b. Voltage of each cell:3.7 V
  c. Total Battery Voltage
     $V_{total} = 3.7 + 3.7$
     $V_{total} = 7.4$ V
  d. Total Energy Capacity
     Energy = Voltage × Capacity
     Energy = 7.4 × 2.5
     Energy = 18.5 Wh
- Current Consumption
  a. ESP32:80 mA
  b. GPS Module:30 mA
  c. GSM SIM800C:350 mA
  d. MPU6050:5 mA
  e. Alcohol Sensor:150 mA
  f. IR Sensor:20 mA
  g. Shock Sensor:5 mA
- Total Current Consumption
  $I_{total} = 80 + 30 + 350 + 5 + 150 + 20 + 5$
  $I_{total} \approx 640$ mA
- Battery Backup Time
  Backup Time = Capacity / Current
  Backup Time = 2500 / 640
  Backup Time ≈ 3.9 hours

2. Power Management Calculation – Bike Unit

- Battery capacity:2000 mAh
  a. Battery voltage:3.7 V
  b. Energy Capacity
  c. Energy = Voltage × Capacity
     Energy = 3.7 V × 2 Ah
     Energy = 7.4 Wh
- Current Consumption
  a. ESP32:80 mA
  b. Relay Module:70 mA
  c. Bluetooth Communication:30 mA
- Total Current Consumption
  Itotal = 80 + 70 + 30
  Itotal = 180 mA
- Battery Backup Time
  Backup Time = Battery Capacity / Total Load Current
  Backup Time = 2000 / 180
  Backup Time ≈ 11 hours

5.Result

5.1Observations/Findings:-

During the development and testing of the Smart Helmet System, several important observations and findings were noted. One of the key observations was the effectiveness of the alcohol detection sensor in identifying alcohol presence from the rider's breath. The sensor successfully detected alcohol levels and prevented the ignition system from starting when alcohol was detected above the preset threshold, improving rider safety. It was also observed that proper calibration of the sensor was necessary to obtain accurate results and avoid false detection.

Additionally, the GPS module effectively captured real-time location data of the rider. However, it required a clear view of the sky to achieve faster satellite connection and accurate positioning. In areas with weak satellite signals, the location update was slightly delayed. The GSM module successfully transmitted emergency messages along with the GPS location to predefined contacts during accident detection, demonstrating reliable communication for emergency situations.

The vibration or tilt sensor used for accident detection responded effectively to sudden impacts or abnormal movements of the helmet. This helped in identifying potential accidents and triggering alert messages. However, it was observed that sensor sensitivity needed proper adjustment to prevent false alarms during sudden braking or rough road conditions.

Overall, the Smart Helmet system demonstrated its ability to enhance road safety by ensuring helmet usage, preventing drunk driving, detecting accidents, and sending emergency alerts with location information. However, proper sensor calibration, stable network connectivity, and efficient power management were found to be essential for reliable long-term performance of the system.

5.2 Result and Discussion: -

The Smart Helmet System provides an effective and reliable solution for improving road safety by integrating various sensors and communication modules. The microcontroller, along with components such as the alcohol sensor, GPS module, GSM module, and accident detection sensor, successfully works together to monitor the rider's condition and ensure safe driving. The system ensures that the vehicle starts only when the rider is wearing the helmet and when no alcohol is detected, thereby reducing the risk of accidents caused by negligence or drunk driving.

One of the most significant outcomes of the system is its ability to provide immediate accident alerts. When an accident occurs, the vibration or tilt sensor detects the sudden impact and activates the alert mechanism. At the same time, the GPS module obtains the rider's exact location and the GSM module sends an emergency message with location details to predefined contacts. This quick notification system helps ensure that emergency assistance can reach the accident location as soon as possible.

Another important feature of the system is real-time location tracking. The GPS module effectively determines the geographical position of the rider and provides accurate location information. However, during testing it was observed that the GPS module requires a strong satellite signal for better accuracy, and open areas provide faster and more reliable location updates compared to indoor or crowded environments.

A key factor affecting the system's performance is network connectivity and sensor calibration. Since the GSM module is responsible for sending alert messages, reliable network coverage is necessary for timely communication. During testing, it was also observed that proper calibration of sensors such as the alcohol sensor and accident detection sensor improved the accuracy of the system and minimized false alerts.

The system was also evaluated for power efficiency and operational reliability. It was found that using a stable power supply and a rechargeable battery helped maintain continuous operation of the helmet system. The overall design of the system ensures low power consumption while maintaining efficient performance for long-term use.

5.3 Output: -

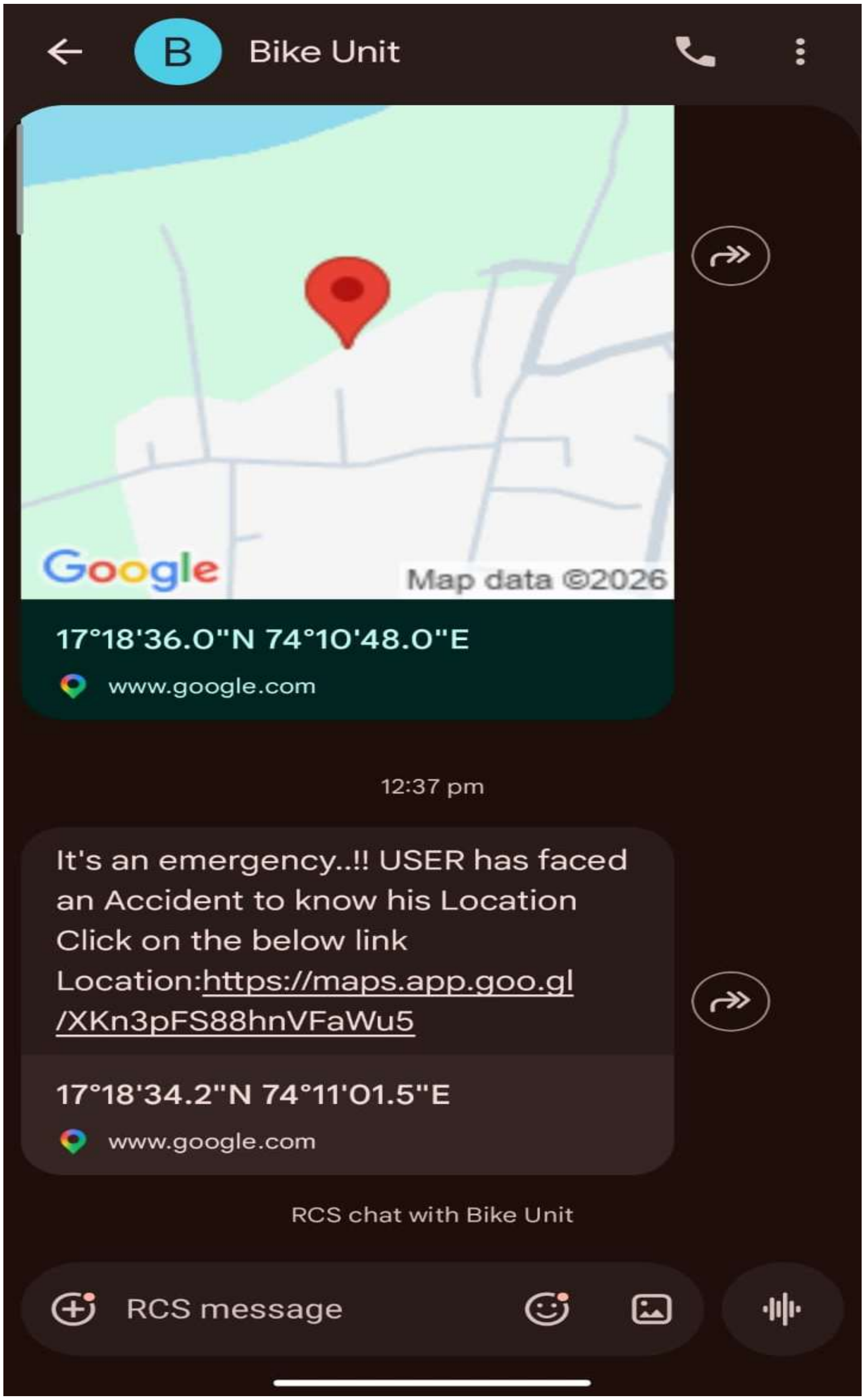


Fig.5.3. Output

6.Appendix

6.1 Flow Chart:-

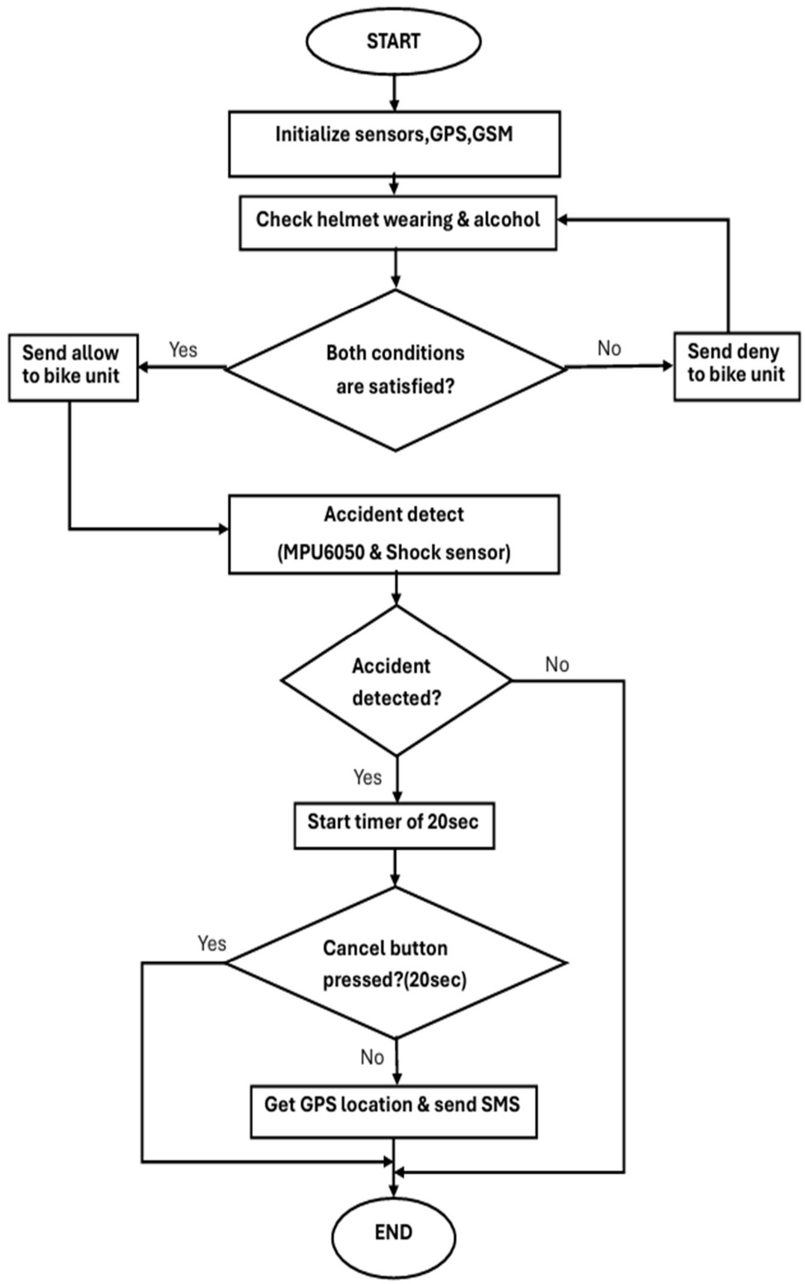


Fig.6.1. Flow Chart

6.2 Algorithm:-

1. Start
2. Initialize System
   - Start Serial communication for debugging.
   - Set IR sensor, shock sensor, and button pins as input.
   - Initialize Bluetooth communication.
   - Initialize GPS module using UART.
   - Initialize GSM module using UART.
   - Initialize MPU6050 accelerometer sensor.
   - Configure GSM module to SMS text mode.
3. Continuous Monitoring (Loop)
   - Repeat the following steps continuously.
4. Helmet Wearing & Alcohol Check

Read IR sensor value to detect if helmet is worn.

- Read alcohol sensor value.
- Compare alcohol value with threshold value.
- Condition:
- If
- Helmet is worn AND
- Alcohol value is below threshold

→ Send "ALLOW" via Bluetooth.

Else

→ Send "DENY" via Bluetooth.

5. Accident Detection
   - Read acceleration values (ax, ay, az) from MPU6050.
   - Calculate tilt angle using acceleration values.
   - Read shock sensor value.
   - Condition:

     If

     Tilt angle > threshold AND

     Shock value > threshold

   → Accident is detected.
   - Store accident detection time.

6. Accident Confirmation
   - Wait for 20 seconds for user response.
   - During this time check cancel button.
   - Condition:

   If button is pressed

   ➞ Cancel accident alert.

   If button is not pressed within 20 seconds

   ➞ Continue to next step.
7. Get GPS Location
   - Read GPS data from GPS module.
   - Decode GPS data using TinyGPS++ library.
   - Extract latitude and longitude.

   If GPS location is received:

   ➞ Proceed to send SMS.

   If GPS location not found:

   ➞ Display GPS Not Found.
8. Send Emergency SMS
   - Send AT command to GSM module.
   - Send SMS to predefined phone number.
   - SMS contains:

     Accident message

     Latitude

     Longitude

     Google Maps link of location
9. End / Continue Monitoring
   - System continues monitoring sensors for the next accident.

6.3 Program / Coding:-

1. Helmet module:

```
#include <WiFi.h>
#include <Wire.h>
#include <MPU6050.h>
#include <TinyGPS++.h>
#include <HardwareSerial.h>
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>

#define IR_SENSOR 32
#define ALCOHOL_SENSOR 34
#define SHOCK_SENSOR 25
#define BUTTON_PIN 12

#define ALCOHOL_THRESHOLD 1600
#define TILT_THRESHOLD 45
#define SHOCK_THRESHOLD 200

MPU6050 mpu;

HardwareSerial gpsSerial(1);
TinyGPSPlus gps;

HardwareSerial gsmSerial(2);

// WiFi router
const char* ssid = "Abdul";
const char* password = "Abdul212 ";

// Bike ESP32 IP
const char* bikeIP = "192.168.1.50";
const int bikePort = 80;

WiFiClient client;

// Telegram
const char* BOTtoken = "YOUR_BOT_TOKEN";
const char* CHAT_ID = "YOUR_CHAT_ID";

WiFiClientSecure telegramClient;
UniversalTelegramBot bot(BOTtoken, telegramClient);

// Authorized SMS number
String authorizedNumber = "+918975545624";

bool accidentDetected = false;
```

```
bool alertSent = false;
unsigned long accidentTime = 0;

// IR state tracking
bool lastIRState = HIGH;
bool currentIRState = HIGH;

////////////////////////////////////////////////////////////

void setup()
{
  Serial.begin(115200);

  pinMode(IR_SENSOR, INPUT);
  pinMode(SHOCK_SENSOR, INPUT);
  pinMode(BUTTON_PIN, INPUT_PULLUP);

  Wire.begin();
  mpu.initialize();

  gpsSerial.begin(9600, SERIAL_8N1, 16, 17);
  gsmSerial.begin(9600, SERIAL_8N1, 26, 27);

  WiFi.begin(ssid, password);

  Serial.print("Connecting WiFi");

  while (WiFi.status() != WL_CONNECTED)
  {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nWiFi Connected");

  telegramClient.setInsecure();

  gsmSerial.println("AT");
  delay(1000);

  gsmSerial.println("AT+CMGF=1");
  delay(1000);

  gsmSerial.println("AT+CNMI=1,2,0,0,0");

  Serial.println("Helmet System Ready");
}
```

```
void loop()
{
  checkHelmetAndAlcohol();

  detectAccident();

  handleAccident();

  checkSMSCancel();

  delay(300);
}

/////////////////////////////////////////////////////////////
// Helmet + Alcohol Check (STATE CHANGE ONLY)


void checkHelmetAndAlcohol()
{
  currentIRState = digitalRead(IR_SENSOR);
  int alcohol = analogRead(ALCOHOL_SENSOR);

  if(currentIRState != lastIRState)
  {
    Serial.println("IR State Changed");

    if(client.connect(bikeIP,bikePort))
    {
      if(currentIRState == LOW && alcohol < ALCOHOL_THRESHOLD)
      {
        client.println("ALLOW");
        Serial.println("Command Sent: ALLOW");
      }
      else
      {
        client.println("DENY");
        Serial.println("Command Sent: DENY");
      }

      client.stop();
    }

    lastIRState = currentIRState;
  }
}
```

```
////////////////////////////////////////////////////////////
// Accident Detection

void detectAccident()
{
  if(accidentDetected) return;

  int16_t ax, ay, az;
  mpu.getAcceleration(&ax,&ay,&az);

  float angleX = atan2((float)ax,(float)az)*180.0/PI;
  float angleY = atan2((float)ay,(float)az)*180.0/PI;

  bool tilt = abs(angleX) > TILT_THRESHOLD ||
              abs(angleY) > TILT_THRESHOLD;

  int shock = analogRead(SHOCK_SENSOR);

  Serial.print("Tilt:");
  Serial.print(tilt);
  Serial.print(" Shock:");
  Serial.println(shock);

  if(tilt && shock > SHOCK_THRESHOLD)
  {
    accidentDetected = true;
    accidentTime = millis();
    alertSent = false;

    Serial.println("Accident Detected!");
  }
}

////////////////////////////////////////////////////////////
// Accident Handling

void handleAccident()
{
  if(!accidentDetected) return;

  if(millis() - accidentTime < 20000)
  {
    Serial.println("Press button or send SMS CANCEL");

    if(digitalRead(BUTTON_PIN) == LOW)
    {
      accidentDetected = false;
      alertSent = false;

      Serial.println("Alert Cancelled by Button");
```

```
    }
  }
  else
  {
    if(!alertSent)
    {
      getGPSAndSendAlert();
      alertSent = true;
    }
  }
}

////////////////////////////////////////////////////////////
// SMS Cancelation

void checkSMSCancel()
{
  if(gsmSerial.available())
  {
    String sms = gsmSerial.readString();
    sms.toUpperCase();

    if(sms.indexOf(authorizedNumber) >= 0 && sms.indexOf("CANCEL") >= 0)
    {
      accidentDetected = false;
      alertSent = false;
      Serial.println("Alert Cancelled by SMS");
    }
  }
}


////////////////////////////////////////////////////////////
// GPS + Alerts

void getGPSAndSendAlert()
{
  float lat = 0;
  float lon = 0;

  unsigned long start = millis();

  while(millis() - start < 10000)
  {
    while(gpsSerial.available())
    {
      gps.encode(gpsSerial.read());

      if(gps.location.isUpdated())
      {
```

```
        lat = gps.location.lat();
        lon = gps.location.lng();

        sendSMS(lat,lon);
        sendTelegram(lat,lon);
        return;
      }
    }
  }
  Serial.println("GPS Not Found");
}

////////////////////////////////////////////////////////////
// SMS Alert

void sendSMS(float lat,float lon)
{
  Serial.println("Sending SMS");
  gsmSerial.print("AT+CMGS=\"+918975545624\"\r");
  delay(1000);

  gsmSerial.println("Emergency! Accident Detected");
  gsmSerial.print("Location: https://maps.google.com/?q=");
  gsmSerial.print(lat);
  gsmSerial.print(",");
  gsmSerial.println(lon);

  gsmSerial.write(26);

  delay(5000);
}

////////////////////////////////////////////////////////////
// Telegram Alert

void sendTelegram(float lat,float lon)
{
  String message="🚨 Accident Detected!\n";

  message+="Location:\nhttps://maps.google.com/?q=";
  message+=String(lat,6);
  message+=",";
  message+=String(lon,6);

  bot.sendMessage(CHAT_ID,message,"");

  Serial.println("Telegram Alert Sent");
}
```

2. Bike module:

```
#include <WiFi.h>
#define RELAY_PIN 2

// Same WiFi router used by helmet
const char* ssid = "Abdul";
const char* password = "Abdul212";

WiFiServer server(80);

String command = "";

void setup()
{
  Serial.begin(115200);

  pinMode(RELAY_PIN, OUTPUT);
  digitalWrite(RELAY_PIN, LOW);

  WiFi.begin(ssid, password);

  Serial.print("Connecting to WiFi");

  while (WiFi.status() != WL_CONNECTED)
  {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nWiFi Connected");

  Serial.print("Bike ESP32 IP Address: ");
  Serial.println(WiFi.localIP());

  server.begin();

  Serial.println("Bike Unit Ready");
}

////////////////////////////////////////////////////////////

void loop()
{
  WiFiClient client = server.available();

  if (client)
  {
    Serial.println("Helmet Connected");
```

```
      command = client.readStringUntil('\n');
      command.trim();

      Serial.print("Command Received: ");
      Serial.println(command);

      if (command == "ALLOW")
      {
        digitalWrite(RELAY_PIN, HIGH);
        Serial.println("Ignition ON");
      }

      else if (command == "DENY")
      {
        digitalWrite(RELAY_PIN, LOW);
        Serial.println("Ignition OFF");
      }

      client.stop();
    }
  }
```

6.4 Advantages / Disadvantages:-

6.4.1 Advantages:-

1. Improved Road Safety:
   - The smart helmet ensures that the rider wears the helmet before starting the vehicle. A sensor placed inside the helmet detects whether it is properly worn. If the helmet is not worn, the system prevents the bike from starting. This helps reduce head injuries and promotes safe riding habits.
2. Prevention of Drunk Driving:
   - The smart helmet uses an alcohol sensor to detect alcohol in the rider's breath. If the alcohol level exceeds the preset limit, the system blocks the ignition of the vehicle. This helps prevent accidents caused by drunk driving and encourages responsible behavior.
3. Automatic Accident Detection:
   - The helmet is equipped with sensors such as vibration or tilt sensors that detect sudden impacts or abnormal movements during an accident. When an accident is detected, the system automatically activates the alert mechanism.
4. Emergency Alert System
   - When an accident occurs, the system sends an emergency message to predefined contacts such as family members or emergency services. This message may include the rider's location and accident information, helping others respond quickly.
5. GPS Location Tracking
   - The GPS module integrated into the smart helmet tracks the real-time location of the rider. In case of an accident or emergency, the exact location can be shared with emergency contacts, making rescue operations faster and easier.
6. Faster Emergency Response
   - Since the system automatically sends alerts with location information, emergency services can reach the accident site more quickly. This can help reduce the severity of injuries and save lives.
7. Encourages Responsible Riding
   - The smart helmet promotes safe driving habits by ensuring helmet usage and preventing drunk driving. Riders become more aware of safety rules, which helps reduce traffic accidents.

8. Low Power Consumption
   - The system uses energy-efficient electronic components such as microcontrollers and sensors. This allows the smart helmet to operate for longer periods without requiring frequent charging or power replacement.
9. Easy to Use and Automatic Operation
   - The smart helmet system operates automatically without requiring manual control from the rider. Once the helmet is worn and the system is activated, it continuously monitors safety conditions.
10. Cost-Effective Safety Technology
   - Compared to advanced vehicle safety systems, the smart helmet project is relatively affordable. It can be implemented using simple electronic components while still providing significant improvements in rider safety.

6.4.2 Disadvantages:-

1. High Initial Cost
- The smart helmet includes several electronic components such as sensors, GPS, GSM modules, and microcontrollers. These components increase the overall cost of the helmet compared to a normal helmet.
2. Dependence on Network Connectivity
- The emergency alert system uses GSM or internet connectivity to send messages. In areas with poor network coverage, the system may not send alerts immediately, which can delay emergency response.
3. GPS Signal Limitations
- The GPS module requires a clear view of the sky to receive satellite signals. In indoor areas, tunnels, or densely populated locations, the GPS signal may be weak or inaccurate.
4. Maintenance and Calibration
- Sensors such as alcohol sensors and accident detection sensors require proper calibration and maintenance. Without regular maintenance, the system may give inaccurate results or false alerts.
5. Possibility of False Alerts
- Sometimes sudden movements, rough roads, or sudden braking may trigger the accident detection sensor even when an accident has not occurred.
6. Increased Weight of Helmet
- Adding electronic components and batteries can increase the weight of the helmet, which may cause discomfort for some riders during long rides.
7. Environmental Sensitivity
- Electronic components inside the helmet may be affected by extreme temperatures, dust, or moisture if not properly protected.
8. Limited Battery Life
- Continuous operation of GPS, GSM, and sensors can consume significant power, which may require frequent charging.
9. Technical Complexity
- The system requires proper design, programming, and integration of multiple components, which can make the system more complex compared to a standard helmt

6.5 Application:-

1. Accident Detection and Alert:
   - Detects accidents using sensors and automatically sends the rider’s location to emergency contacts.
2. Helmet Detection for Bike Start:
   - The bike will start only when the rider wears the helmet, improving safety.
3. Alcohol Detection:
   - Detects alcohol using an alcohol sensor and prevents the rider from starting the vehicle.
4. GPS Tracking:
   - Sends the rider’s real-time location using GPS and GSM modules.
5. Emergency Notification:
   - Sends an SMS to family or emergency services during accidents.
6. Worker Safety Monitoring:
   - Used in construction sites or mines to monitor worker safety.
7. Navigation Assistance:
   - Provides location and navigation support using GPS

7.Conclusion

7.1 Future Scope:-

1. Enhanced Sensor Integration
   - Future versions of the smart helmet can integrate advanced sensors such as gas sensors, proximity sensors, and fatigue detection sensors to improve rider safety.
   - Additional temperature and environmental sensors can monitor surrounding conditions and warn riders about extreme weather or hazardous environments.
2. Machine Learning–Based Predictive Analysis
   - Artificial Intelligence (AI) and machine learning algorithms can analyze rider behavior patterns to detect fatigue, distraction, or unsafe driving habits.
   - The system can also predict potential accidents by analyzing speed, tilt angle, and motion data from sensors such as accelerometers and gyroscopes.
3. Cloud-Based Monitoring System
   - The smart helmet can connect to cloud platforms for real-time monitoring and data storage.
   - Accident data and ride history can be stored in the cloud, enabling data analysis and improved safety insights.
4. IoT Integration for Smart Transportation
   - The helmet can be connected to smartphone applications to provide navigation alerts, emergency notifications, and ride statistics.
   - Integration with smart traffic systems and connected vehicles can help provide warnings about road conditions, traffic congestion, or nearby accidents.
5. Automated Safety Mechanisms
   - Future versions can include automatic accident detection systems that send emergency alerts with GPS location to family members or emergency services.
   - Integration with the motorcycle's ignition system can ensure that the bike starts only when the rider is wearing the helmet, improving compliance with safety rules.

6. Wearable Health Monitoring
   - Smart helmets can integrate health monitoring sensors to track the rider's heart rate, stress levels, or fatigue.
   - In case of abnormal health conditions or accidents, the system can automatically trigger emergency alerts.
7. Improved Power Efficiency and Backup Solutions
   - Using low-power microcontrollers and energy-efficient sensors can increase battery life.
   - Adding solar panels or improved rechargeable battery systems can ensure longer operational time for the helmet.
8. Scalability and Smart City Integration
   - Smart helmets can be connected to smart city infrastructure to enhance road safety and traffic monitoring.
   - Large-scale deployment can help authorities analyze accident-prone zones and improve traffic management systems.
9. Global Adoption and Safety Standardization
   - With improved design and safety features, smart helmets can become a standard safety device for two-wheeler riders worldwide.
   - Compliance with international road safety and electronic device standards will make the system suitable for commercial production and widespread adoption.

7.2 Conclusions

In conclusion, the Smart Helmet System provides an effective and reliable solution for enhancing road safety and reducing accidents caused by negligence and drunk driving. This project successfully integrates the ESP32-microcontroller, SIM800C GSM module, NEO-6M GPS module, IR sensor, MQ3 alcohol sensor, MPU6050 gyroscope and accelerometer, and shock or SW-420 vibration sensor to monitor rider safety and detect accidents in real time. The system ensures that the rider wears the helmet and is not under the influence of alcohol before the bike can start, thereby promoting responsible and safe driving behavior.

The ESP32 microcontroller acts as the central processing unit, coordinating all sensors and communication modules in the helmet unit and bike unit. The IR sensor detects whether the helmet is worn properly, while the MQ3 alcohol sensor checks the rider's breath for alcohol presence. If alcohol is detected or the helmet is not worn, the system prevents the bike ignition through the relay module in the bike unit. Additionally, the MPU6050 sensor and shock sensor help detect sudden impacts or abnormal movements that may indicate an accident.

In the event of an accident, the system quickly gathers location data using the NEO-6M GPS module and sends an emergency alert message through the SIM800C GSM module to predefined contacts. This feature enables faster emergency response and improves the chances of timely medical assistance. The push button switch can also be used by the rider to manually send emergency alerts if required.

The system is powered using a 3.7V 18650 lithium-ion battery with a capacity of 2600mAh, along with a TP4056 charging module for safe battery charging. Power regulation is achieved using DC-DC converters such as the MT3608 step-up converter and LM2596 step-down converter to provide stable voltage levels to different components. Additional components like resistors, capacitors, GSM antenna, and connecting wires ensure stable circuit operation and reliable communication.

During the testing phase, the system demonstrated reliable performance in detecting helmet usage, alcohol presence, and potential accidents while maintaining efficient power consumption.

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- Research Paper

International Research Journal of Engineering and Technology (IRJET) e-ISSN: 2395-0056
Volume: 08 Issue: 04 | Apr 2021 www.irjet.net p-ISSN: 2395-0072

Smart Helmet for Accident Prevention and Detection

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Abstract - *The objective of the smart helmet is to provide a means and apparatus for detecting and reporting accidents. Sensors, and cloud computing infrastructures are utilised for building the system. The accident detection system communicates the accelerometer values to the processor which continuously monitors for erratic variations. When an accident occurs, the related details are sent to the emergency contacts by utilizing a cloud based service. The vehicle location is obtained by making use of the global positioning system. The system promises a reliable and quick delivery of information relating to the accident in real time and up dated to cloud which are accessed by IOT. Thus, by making use of the ubiquitous connectivity which is a salient feature for the smart cities, a smart helmet for accident detection is built.*

Key Words: smart helmet, accident detection, Arduino, IoT, Sensor

1. INTRODUCTION

India has a huge number of road accidents every year. The accidents may be due to many reasons like by drink and drive, driving rashly, exceeding the speed limit, etc. Sometimes, the person who gets injured might not be responsible for the accident. It might be the fault of some other vehicle rider. But overall both riders will get affected. Due to a lack of first aid and emergency medical services on time, the riders may die. Some deaths are due to the ambulance not reaching the desired location on time. In case of an accident, to save time and inform the concerned person, a system is proposed which can make sure that the rider gets the required attention in a short time. In India, many people use two-wheeler vehicles as compared to four-wheeler vehicles because of its low cost and simplicity. In many accidents, the rider gets injured mainly on the head. A helmet plays a very important role in saving the life of the ridden So to encourage people to wear helmets and to avoid accidents, a design is proposed that synchronizes the module present in bike.

2. LITRETURE REVIEW

[1]. Hussain A.Attia1, Shereen Ismail enhanced electronic safety system design with simulation results for teenagers and older drivers is presented in this paper. Because of their physiological characteristics that lead to multiple driving errors, which need monitoring to avoid their recurrence. Comparing to the initial design, the presented safety system in this study considers additional two parameters; the number of driving errors and the errors duration. Based on these two parameters, the total number of recorded driving errors (lower/higher than the low/high front distance limits respectively) will be considered. If this number exceeds a certain limit of error then a suitable response will be taken as a safety reaction. Simulation results are demonstrating the recognition capability among the three cases of driving conditions, which are safe front distance, short front distance alarm, and long front distance alarm. In addition, the results are reflecting the highly effectiveness of the system in term of response and promising the possibility of obtaining high performance system in the fields of driving safety.

[2]. RistoÖörni talk about the demand for four intelligent vehicle safety systems (IVSSs) – emergency braking, speed alert, blind spot monitoring and lane keeping support – is analysed by constructing their demand curves (demand as a function of product price) based on data available from user interviews and a literature study. The study also provides a method for constructing linear and exponential demand curves of the systems from data gathered from user interviews. The estimated linear and exponential demand curves were tested by least-squares fitting to the data collected from user interviews. The mean absolute error was consistently larger for all of the systems studied here when using the linear instead of exponential model. This suggests that the exponential model reflects more accurately the demand for IVSSs than does the linear model.

[3]. Shouvik Chakraborty, Sachidananda Sen explained about an increasing number of accidents of Vehicles has led to the study and design of Active safety systems in modern automobiles. For the purpose, a number of sensors for the measurement of Vehicle yaw, wheel velocities and acceleration are deployed. However, some key parameters like slip angle and frictional forces are hard to be determined using sensors and also cost prohibitive. Estimation of friction coefficient and frictional forces has been of wide importance for the design of Active safety systems as the information is required for the design of efficient control system. Besides, the system being highly nonlinear in nature, linearized

estimation technique may lead to high approximation errors. This paper presents an unscented Kalman Filter based estimation algorithm for a specific nonlinear tire model for the estimation of friction coefficient and lateral and longitudinal frictional forces.

[4]. Chuchu Fan, Bolun Qi published the paper about the Safety analysis of Autonomous Vehicles and Advanced Driver Assist Systems (ADAS) is a central challenge facing the automotive industry. In this paper, we present a recently developed data-driven formal verification technique and demonstrate its applicability in a case study involving integrated safety analysis of an Automatic Emergency Braking (AEB) system. Our technique combines model-based, hybrid system reachability analysis with sensitivity analysis of components with possibly unknown or inaccessible models. The scenarios we consider for safety analysis are representative of the most common type of rear-end crashes, which are used for evaluating AEB and forward collision avoidance systems. We show that our verification tool Dry VR can effectively establish safety of these scenarios (specified by parameters like braking profiles, initial velocities, uncertainties in position and reaction times), and compute the severity of accidents for unsafe scenarios. The analysis can quantify the safety envelope of the system in the parameter space which is valuable for both design and certification. We also show how the reachability analysis can be combined with statistical information about the parameters, to assess the risk-level of the system, which in turn is essential, for determining Automotive Safety Integrity Levels (ASIL) mandated by the ISO26262 standard.

[5]. Mallikarjuna Gowda C P, Raju Hajare talk about the proposed system aims at developing and designing a suitable system for automobile purposes using Zigbee protocols. The main problems faced in the existing system are inaccuracies in the calculation of speed, distance measurement, and slow response time, etc. The proposed system solves many of the problems faced by the existing systems by using a GPS module instead of the conventional speedometer and also uses sensors which are reliable in areas where human intervention is either unintended or where it puts life to risk. The problems of traffic congestion in urban arterials are increasing day by day and it is very difficult to handle it during emergencies. So we are developing a communication unit within the system to interact with other vehicles in order to clear the lanes. This system aims at communicating with the vehicle in its surrounding with the help of its location (i.e., using the latitude and longitude) to indicate their proximity. When these vehicles are very close in proximity the drivers are cautioned with the help of a message. In this way the drivers can communicate with each other and act according to the situation.

3. Methodology

3.1 Arduino UNO

The UNO is the best board to get started with electronics and coding. If this is your first experience tinkering with the platform, the UNO is the most robust board you can start playing with. The UNO is the most used and documented board of the whole Arduino family. Arduino Uno is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz quartz crystal, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. You can tinker with your UNO without worrying too much about doing something wrong, worst case scenario you can replace the chip for a few dollars and start over again.

Fig -1: Arduino UNO

Fig.1. reprsents the Arduino UNO Chip where Arduino is open-source hardware. The hardware reference designs are distributed under a Creative Commons Attribution Share-Alike 2.5 license and are available on the Arduino website. Layout and production files for some versions of the hardware are also available.

3.2 Gyroscope Sensor

The ADXL335 is a small, thin, low power, complete 3-axis accelerometer with signal conditioned voltage outputs. The product measures acceleration with a minimum full-scale range of ±3 *g*. It can measure the static acceleration of gravity in tilt sensing applications, as well as dynamic acceleration resulting from motion, shock, or vibration. The user selects the bandwidth of the accelerometer using the CX, CY, and CZ capacitors at the XOUT, YOUT, and ZOUT pins. Bandwidths can be selected to suit the application, with a range of 0.5 Hz to 1600 Hz for X and Y axes, and a range of 0.5 Hz to 550 Hz for the Z axis. The ADXL335 is available in a small, low profile, 4 mm × 4 mm × 1.45 mm, 16-lead, plastic lead frame chip scale package(LFCSP_LQ).

Fig -2: Gyroscope Sensor

The ADXL335 is a complete 3-axis acceleration measurement system. The ADXL335 has a measurement range of ±3 *g* minimum. It contains a polysilicon surface micro machined sensor and signal conditioning circuitry to implement an openloop acceleration measurement architecture. The output signals are analog voltages that are proportional to acceleration. The accelerometer can measure the static acceleration of gravity in tilt sensing applications as well as dynamic acceleration resulting from motion, shock, or vibration. The sensor is a poly silicon surface micro machined structure built on top of a silicon wafer. Polysilicon springs suspend the structure over the surface of the wafer and provide a resistance against acceleration forces. Deflection of the structure is measured using a differential capacitor that consists of independent fixed plates and plates attached to the moving mass. The fixed plates are driven by 180° out-of-phase square waves. Acceleration deflects the moving mass and unbalances the differential capacitor resulting in a sensor output whose amplitude is proportional to acceleration. Phase-sensitive demodulation techniques are then used to determine the magnitude and direction of the acceleration. The demodulator output is amplified and brought off-chip through a 32 kΩ resistor. The user then sets the signal bandwidth of the device by adding a capacitor. This filtering improves measurement resolution and helps prevent aliasing.

3.3 Touch Sensor

Our senses inform to us when our hands touch something. Computer input devices are indifferent to human contact as there is no reaction from software in the event of The sense of touch is an important sensory channel in many animals and some plants. making, maintaining or breaking physical contact like touches or releases. Thus, touch sensing input devices offers numerous possibilities for novel interaction techniques. Touch sensor technology is slowly replacing the mechanical objects like mouse and keyboard. A touch sensor detects touch or near proximity without relying on physical contact. Touch sensors are making their way into many applications like mobile phones, remote controls, and control panels, etc. Present day touch sensors can replace mechanical buttons and switches.

Fig -3: Touch Sensor

Touch sensors with simple rotational sliders, touch pads and rotary wheels offer significant advantages for more intuitive user interfaces. Touch sensors are more convenient and more reliable to use without moving parts. The use of touch sensors provides great freedom to the system designer and help in reducing the overall cost of the system. The overall look of the system can be more appealing and contemporary.

3.4 Vibration Sensor

Vibration Sensor is a high sensitivity non-directional vibration sensor. When the module is stable, the circuit is turned on and the output is high. When the movement or vibration occurs, the circuit will be briefly disconnected and output low. At the same time, you can also adjust the sensitivity according to your own needs. The vibration switch that opens when vibration is detected and closes when there is no vibration.

Fig -4: Touch Sensor

3.5 Gas Sensor

Sensitive material of MQ-2 gas sensor is SnO2, which with lower conductivity in clean air. When the target combustible gas exist, The sensor's conductivity is more higher along with the gas concentration rising. Please use simple electro circuit, Convert change of conductivity to correspond output signal of gas concentration. MQ-2 gas sensor has high sensitivity to LPG, Propane and Hydrogen, also could be used to Methane and other combustible steam, it is with low cost and suitable for different application. Sensor is sensitive to flammable gas and smoke. Smoke sensor is given 5 volt to power it. Smoke sensor indicate smoke by the voltage that it outputs .More smoke more output. A potentiometer is provided to adjust the sensitivity. But when smoke exist sensor provides an analog resistive output based on concentration of smoke. The circuit has a heater. Power is given to heater by VCC and GND from power supply. The circuit has a variable resistor. The resistance across the pin depends on the smoke in air in the sensor. The resistance will be lowered if the content is more. And voltage is increased between the sensor and load resistor.

Fig -5: Gas Sensor

3.6 Relay

Relays are the primary protection as well as switching devices in most of the control processes or equipment. All the relays respond to one or more electrical quantities like voltage or current such that they open or close the contacts or circuits. A relay is a switching device as it works to isolate or change the state of an electric circuit from one state to another.

Fig -6: Relay

Classification or the types of relays depend on the function for which they are used. Some of the categories include protective, reclosing, regulating, auxiliary and monitoring relays. Protective relays continuously monitor these parameters: voltage, current, and power; and if these parameters violate from set limits they generate alarm or isolate that particular circuit. These types of relays are used to protect equipment like motors, generators, and transformers, and so on. Reclosing relays are used to connect various components and devices within the system network, such as synchronizing process, and to restore the various devices soon after any electrical fault vanishes, and then to connect transformers and feeders to line network. Regulating relays are the switches that contacts such that voltage boosts up as in the case of tap changing transformers.

3.7 Direct Current (DC) motor

Almost every mechanical development that we see around us is accomplished by an electric motor. Electric machines are a method of converting energy. Motors take electrical energy and produce mechanical energy. Electric motors are utilized to power hundreds of devices we use in everyday life. Electric motors are broadly classified into two different categories: Direct Current (DC) motor and Alternating Current (AC) motor. In this article we are going to discuss about the DC motor and its working. And also how a gear DC motors works. A DC motor is an electric motor that runs on direct current power. In any electric motor, operation is dependent upon simple electromagnetism. A current carrying conductor generates a magnetic field, when this is then placed in an external magnetic field, it will encounter a force proportional to the current in the conductor and to the strength of the external magnetic field. It is a device which converts electrical energy to mechanical energy. It works on the fact that a current carrying conductor placed in a magnetic field experiences a force which causes it to rotate with respect to its original position.

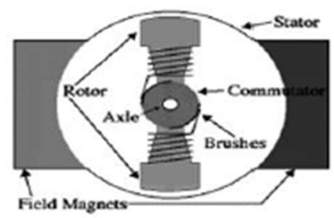

Fig -7: DC Motor

The rotor consists of windings, the windings being electrically associated with the commutator. The geometry of the brushes, commutator contacts and rotor windings are such that when power is applied, the polarities of the energized winding and the stator magnets are misaligned and the rotor will turn until it is very nearly straightened with the stator's field magnets.

3.8 Liquid Crystal Display

LCD screen is an electronic display module and find a wide range of applications. A 16x2 LCD display is very basic module and is very commonly used in various devices and circuits.

A **16x2 LCD** means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers, namely, Command and Data. The command register stores the command instructions given to the LCD. A command is an instruction given to LCD to do a predefined task like initializing it, clearing its screen, setting the cursor position, controlling display etc. The data register stores the data to be displayed on the LCD. The data is the ASCII value of the character to be displayed on the LCD.

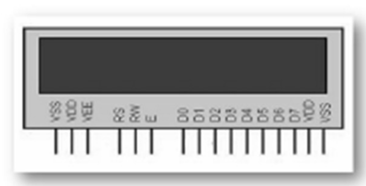

Fig -8: Pin Diagram of LCD

3.9 Global Positioning System

GPS or Global Positioning System is a satellite navigation system that furnishes location and time information in all climate conditions to the user. GPS is used for navigation in planes, ships, cars and trucks also. The system gives critical abilities to military and civilian users around the globe. GPS provides continuous real time, 3-dimensional positioning, navigation and timing worldwide. The Global Positioning System (GPS) is a satellite-based navigation system made up of at least 24 satellites. GPS works in any weather conditions, anywhere in the world, 24 hours a day, with no subscription fees or setup charges. The U.S. Department of Defense (USDOD) originally put the satellites into orbit for military use, but they were made available for civilian use in the 1980s.

Fig -9: Global Positioning System

3.10 Global System for Mobile Communication

Global system for mobile communication (GSM) is a globally accepted standard for digital cellular communication. GSM is the name of a standardization group established in 1982 to create a common European mobile telephone standard that would formulate specifications for a pan-European mobile cellular radio system operating at 900 MHz. It is estimated that many countries outside of Europe will join the GSM partnership.

Fig -10: GSM

GSM/GPRS Modem-RS232 is built with Dual Band GSM/GPRS engine- SIM900, works on frequencies 900/ 1800 MHz. The Modem is coming with RS232 interface, which allows you connect PC as well as microcontroller with RS232 Chip(MAX232). The baud rate is configurable from 9600-115200 through AT command. The GSM/GPRS Modem is having internal TCP/IP stack to enable you to connect with internet via GPRS. It is suitable for SMS, Voice as well as DATA transfer application in M2M interface. The onboard Regulated Power supply allows you to connect wide range unregulated power supply . Using this modem, you can make audio calls, SMS, Read SMS, attend the incoming calls and internet through simple AT commands.

3.11 Zigbee Protocol

The NRF24L01+ is a single chip 2.4GHz transceiver with an embedded baseband protocol engine (Enhanced ShockBurst™), suitable for ultra-low power wireless applications. The nRF24L01+ is designed for operation in the world wide ISM frequency band at 2.400 - 2.4835GHz. To design a radio system with the nRF24L01+, you simply need an MCU (microcontroller) and a few external passive components. You can operate and configure the nRF24L01+ through a Serial Peripheral Interface (SPI). The register map, which is accessible through the SPI, contains all configuration registers in the nRF24L01+ and is accessible in all operation modes of the chip. The embedded baseband protocol engine (Enhanced ShockBurst) is based on packet communication and supports various modes from manual operation to advanced autonomous protocol operation. Internal FIFOs ensure a smooth data flow between the radio front end and the system's MCU. Enhanced Shock- Burst reduces system cost by handling all the high speed link layer operations.

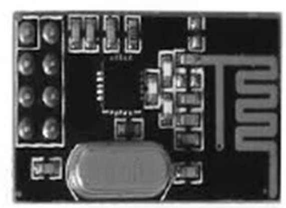

Fig -11: Zigbee Protocol

The radio front end uses GFSK modulation. It has user configurable parameters like frequency channel, output power and air data rate. NRF24L01+ supports an air data rate of 250 kbps, 1 Mbps and 2Mbps. The high air data rate combined with two powers saving modes make the nRF24L01+ very suitable for ultra-low power designs. NRF24L01+ is drop-in compatible with nRF24L01 and on-air compatible with nRF2401A, nRF2402, nRF24E1 and nRF24E2. Intermodulation and wideband blocking values in nRF24L01+ are much improved in comparison to the nRF24L01 and the addition of internal filtering to nRF24L01+ has improved the margins for meeting RF regulatory standards.

4. FUNCTIONALITY

In this system, we use ARDUINO UNO (ATmega328P) microcontroller which acts as brain of the system, because the entire system program instruction stored in it. Here we have two section in which one is at the helmet and the other is at the vehicle. The helmet section has touch sensor which

International Research Journal of Engineering and Technology (IRJET) e-ISSN: 2395-0056
Volume: 08 Issue: 04 | Apr 2021 www.irjet.net p-ISSN: 2395-0072

gives the information about the status of whether the helmet is occupied or not, also we have sensors like gyroscope and vibration sensor to know the status of accident. The location of accident area along SMS or either call have done by the GSM module we have. All the data fetched from the helmet section is transferred through RF technology and reception take place at the vehicle section. The data is also update to cloud so that we are able to either control or monitor the respective system using IOT.

Fig -12: Algorithm of Smart Helmet

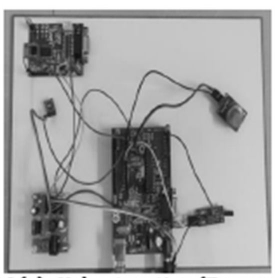
Fig -13(a): Helmet section (Transmitter)

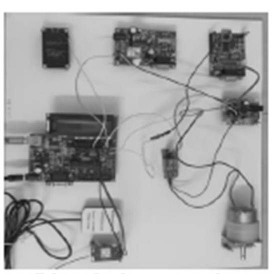
Fig -13(b): Vehicle section (Receiver)

It is already mentioned that the project is divided into two units namely helmet and bike. In helmet unit, also called the transmitter unit shown in Fig.13(a),the force sensing resistor is placed on inside upper part of the helmet where actually head will touch with sensor surface. And alcohol sensor is placed on in front of rider's mouth so that it can sense easily. And the battery and regular circuits were fixed inside the helmet. Secondary controller and RF transmitter circuit were also placed inside the helmet. Antenna is located outside the helmet. The receiver unit shown in Fig.13(b) is placed in the bike. The RF receiver accepts all the data from the helmet (i.e transmitter) unit. Depending on the conditions, if true, the ignition starts and bike moves. The GSM can continuously send the location information of the bike. If any accident occurs, the vibration sensor gets activated and sends the location information to the registered mobile number.

5. FUTURE IMPROVEMENTS

The above mentioned solutions are either dependent on some hardware such as sensors that have to be present in the car or require a smartphone to be present within the vehicle. Although the use of such hardware turns out to be a more cost-efficient approach, it has the drawback of being destroyed in the accident and hence giving false or no readings at all. Therefore, a competent solution that does not depend on any hardware device or sensor is required for the prevention of traffic accidents. Further improvisations include installing a vision system for recording the activities of the driver. The recorded information then can be used by the controlling authority for monitoring the traffic and safety rules. It can be upgraded by mounting the wireless transmitter on cars which is helpful for enhanced communication vehicle to vehicle.

6. CONCLUSION

Road accidents are increasing day by day because the riders are not using the helmet and due to consumption of alcohol. In today's world, huge numbers of people are dying on road accidents. By using smart helmet, the accidents can be detected. The main target of the project is designing a smart helmet for accident avoidance and alcohol detection. The IR sensor checks if the person is wearing the helmet or not. The Gas sensor recognizes the alcoholic substance in the rider's breath. If the person is not wearing the helmet and if he consumes alcohol , the bike will not start. If there is no sign of alcoholic substance present and helmet is used, then only the bike will start. At the point when the rider met with an accident, the sensor recognizes the condition of the motorbike and reports the accident. Then the GPS in the bike will send the location of the accident place to main server of the nearby hospitals.

7. REFERENCES

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- Total bill of Materials:

Sr.No.	Component	Quantity	Cost per unit	Total cost
1.	Esp 32	2	350/-	700/-
2.	GSM 800C	1	950/-	950/-
3.	GPS NEO-6m	1	350/-	350/-
4.	IR senor	1	35/-	35/-
5.	MQ3	1	150/-	150/-
4.	MPU6050	1	290/-	290/-
5.	Shock sensor	1	50/-	50/-
6.	DC-DC boost converter	2	40/-	80/-
7.	DC-DC buck converter	2	60/-	120/-
8.	Relay module	1	40/-	40/-
9.	Push button	1	10/-	10/-
10.	Resistor	3	5/-	15/-
11.	Capacitor	1	10/-	10/-
12.	PCB	3	50/-	150/-
13.	18650 lithium ion battery	3	80/-	240/-
14.	TP4056 charging module	2	45/-	90/-
15.	Jumper wires(M-M,M-F)	50	5/-	250/-
16.	Header pins	4(set)	15/-	60/-
17.	Single strand connecting wires	10(meter)	10/-(per meter)	100/-
18.	Helmet	1	500/-	500/-
19.	Antenna	1	180/-	180/-
20.	DC Motor with wheel	1	50/-	50/-
21.	Battery holder	2	50/-	100/-
22.	SIM	1	50/-	50/-
23.	Recharge	-	250/-	250/-
24.	Others	-	-	400/-
	Total			5220/-

- Datasheets:-

Table 1: Pin Description

Name	No.	Type	Function
			Analog
VDDA	1	P	Analog power supply (2.3 V ~ 3.6 V)
LNA_IN	2	I/O	RF input and output
VDD3P3	3	P	Analog power supply (2.3 V ~ 3.6 V)
VDD3P3	4	P	Analog power supply (2.3 V ~ 3.6 V)
			VDD3P3_RTC
SENSOR_VP	5	I	GPIO36, ADC1_CH0, RTC_GPIO0
SENSOR_CAPP	6	I	GPIO37, ADC1_CH1, RTC_GPIO1
SENSOR_CAPN	7	I	GPIO38, ADC1_CH2, RTC_GPIO2
SENSOR_VN	8	I	GPIO39, ADC1_CH3, RTC_GPIO3
CHIP_PU	9	I	High: On; enables the chip Low: Off; the chip powers off Note: Do not leave the CHIP_PU pin floating.
VDET_1	10	I	GPIO34, ADC1_CH6, RTC_GPIO4
VDET_2	11	I	GPIO35, ADC1_CH7, RTC_GPIO5
32K_XP	12	I/O	GPIO32, ADC1_CH4, RTC_GPIO9, TOUCH9, 32K_XP (32.768 kHz crystal oscillator input)
32K_XN	13	I/O	GPIO33, ADC1_CH5, RTC_GPIO8, TOUCH8, 32K_XN (32.768 kHz crystal oscillator output)
GPIO25	14	I/O	GPIO25, ADC2_CH8, RTC_GPIO6, DAC_1, EMAC_RXD0
GPIO26	15	I/O	GPIO26, ADC2_CH9, RTC_GPIO7, DAC_2, EMAC_RXD1
GPIO27	16	I/O	GPIO27, ADC2_CH7, RTC_GPIO17, TOUCH7, EMAC_RX_DV
MTMS	17	I/O	GPIO14, ADC2_CH6, RTC_GPIO16, TOUCH6, EMAC_TXD2, HSPICLK, HS2_CLK, SD_CLK, MTMS
MTDI	18	I/O	GPIO12, ADC2_CH5, RTC_GPIO15, TOUCH5, EMAC_TXD3, HSPIQ, HS2_DATA2, SD_DATA2, MTDI
VDD3P3_RTC	19	P	Input power supply for RTC IO (2.3 V ~ 3.6 V)
MTCK	20	I/O	GPIO13, ADC2_CH4, RTC_GPIO14, TOUCH4, EMAC_RX_ER, HSPID, HS2_DATA3, SD_DATA3, MTCK
MTDO	21	I/O	GPIO15, ADC2_CH3, RTC_GPIO13, TOUCH3, EMAC_RXD3, HSPICS0, HS2_CMD, SD_CMD, MTDO

[redacted]

Name	No.	Type	Function
GPIO2	22	I/O	GPIO2, ADC2_CH2, RTC_GPIO12, TOUCH2, HSPIWP, HS2_DATA0, SD_DATA0
GPIO0	23	I/O	GPIO0, ADC2_CH1, RTC_GPIO11, TOUCH1, EMAC_TX_CLK, CLK_OUT1,
GPIO4	24	I/O	GPIO4, ADC2_CH0, RTC_GPIO10, TOUCH0, EMAC_TX_ER, HSPIHD, HS2_DATA1, SD_DATA1
			VDD_SDIO
GPIO16	25	I/O	GPIO16, HS1_DATA4, U2RXD, EMAC_CLK_OUT
VDD_SDIO	26	P	Output power supply: 1.8 V or the same voltage as VDD3P3_RTC
GPIO17	27	I/O	GPIO17, HS1_DATA5, U2TXD, EMAC_CLK_OUT_180
SD_DATA_2	28	I/O	GPIO9, HS1_DATA2, U1RXD, SD_DATA2, SPIHD
SD_DATA_3	29	I/O	GPIO10, HS1_DATA3, U1TXD, SD_DATA3, SPIWP
SD_CMD	30	I/O	GPIO11, HS1_CMD, U1RTS, SD_CMD, SPICS0
SD_CLK	31	I/O	GPIO6, HS1_CLK, U1CTS, SD_CLK, SPICLK
SD_DATA_0	32	I/O	GPIO7, HS1_DATA0, U2RTS, SD_DATA0, SPIQ
SD_DATA_1	33	I/O	GPIO8, HS1_DATA1, U2CTS, SD_DATA1, SPID
			VDD3P3_CPU
GPIO5	34	I/O	GPIO5, HS1_DATA6, VSPICS0, EMAC_RX_CLK
GPIO18	35	I/O	GPIO18, HS1_DATA7, VSPICLK
GPIO23	36	I/O	GPIO23, HS1_STROBE, VSPID
VDD3P3_CPU	37	P	Input power supply for CPU IO (1.8 V ~ 3.6 V)
GPIO19	38	I/O	GPIO19, U0CTS, VSPIQ, EMAC_TXD0
GPIO22	39	I/O	GPIO22, U0RTS, VSPIWP, EMAC_TXD1
U0RXD	40	I/O	GPIO3, U0RXD, CLK_OUT2
U0TXD	41	I/O	GPIO1, U0TXD, CLK_OUT3, EMAC_RXD2
GPIO21	42	I/O	GPIO21, VSPIHD, EMAC_TX_EN
			Analog
VDDA	43	P	Analog power supply (2.3 V ~ 3.6 V)
XTAL_N	44	O	External crystal output
XTAL_P	45	I	External crystal input
VDDA	46	P	Analog power supply (2.3 V ~ 3.6 V)
CAP2	47	I	Connects to a 3.3 nF (10%) capacitor and 20 kΩ resistor in parallel to CAP1

[redacted]

Name	No.	Type	Function
CAP1	48	I	Connects to a 10 nF series capacitor to ground
GND	49	P	Ground

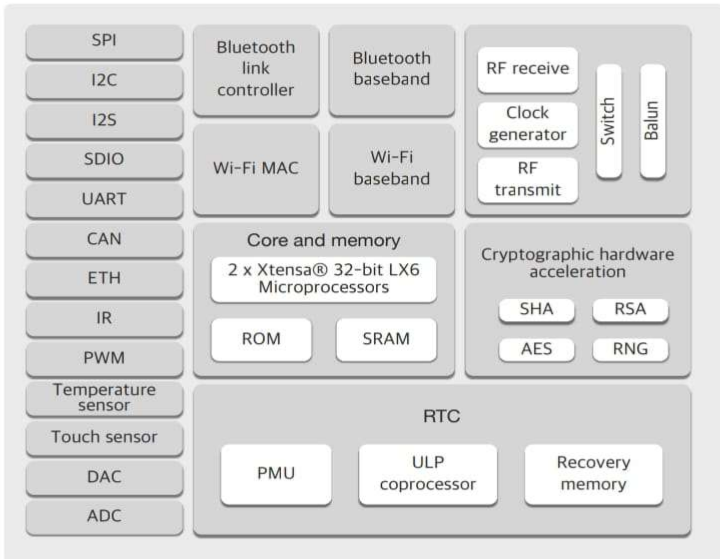

Figure 1: Function Block Diagram

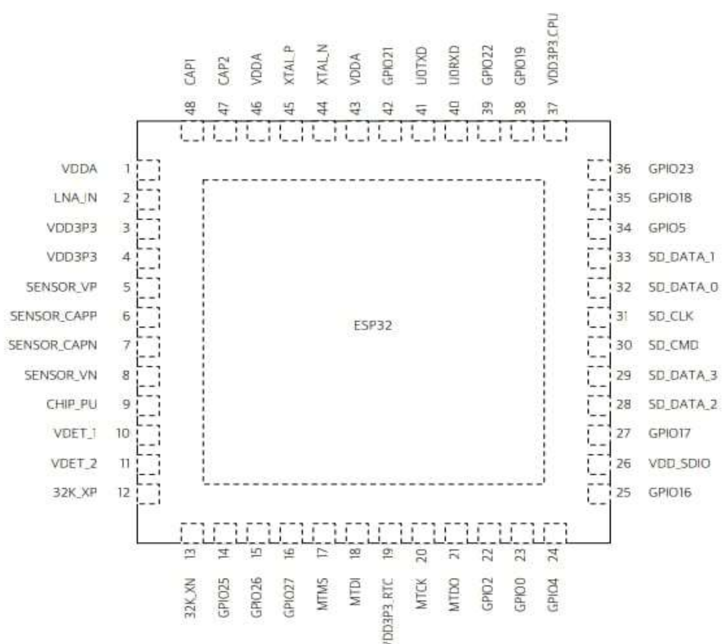

Figure 2: ESP32 Pin Layout

SIM800C Module

SIM 800C Module is a complete Quad-band GSM/GPRS solution in a SMT type, which can be embedded in the customer applications.

These modules are sub-system of the Internet-of-everything hardware. SIM800C supports Quad-band 850/900/1800/1900MHz, it can transmit Voice, SMS and data information with low power consumption. With tiny size of 17.6*15.7*2.3mm, it can smoothly fit into slim and compact demands of customer design.

SPECIFICATIONS:

GSM	850,900,1800 and 1900MHz
FLASH	SIM800C (24Mbit) SIM800C32 (32Mbit)
RAM	32Mbit
Power supply	3.4V ~4.4V
Power saving	Typical power consumption in sleep mode is 0.88mA (BS-PA-MFRMS=9)

Frequency bands	Quad-band: GSM 850, EGSM 900, DCS 1800, PCS 1900. SIM800C can search the 4 frequency bands automatically. The frequency bands can also be set by AT command "AT+CBAND".
	Compliant to GSM Phase 2/2+
Transmitting power	Class 4 (2W) at GSM 850 and EGSM 900
	Class 1 (1W) at DCS 1800 and PCS 1900
GPRS connectivity	GPRS multi-slot class 12（default）
	GPRS multi-slot class 1~12 (option)
Temperature range	Normal operation: -40°C ~ +85°C
	Storage temperature -45°C ~ +90°C
Data GPRS	GPRS data downlink transfer: max. 85.6 kbps
	GPRS data uplink transfer: max. 85.6 kbps
	Coding scheme: CS-1, CS-2, CS-3 and CS-4
	PAP protocol for PPP connect
	Integrate the TCP/IP protocol.
	Support Packet Broadcast Control Channel (PBCCH)
USSD	Unstructured Supplementary Services Data (USSD) support
SMS	MT, MO, CB, Text and PDU mode
	SMS storage: SIM card

Receiver performance data

Receiver type	50-channel u-blox 6 engine GPS L1 C/A code SBAS: WAAS, EGNOS, MSAS		
Navigation update rate	up to 5 Hz		
Accuracy[1]	Position SBAS	2.5 m CEP 2.0 m CEP	
Acquisition[1]		NEO-6G/Q	NEO-6M
	Cold starts: Aided starts[2]: Hot starts:	26 s 1 s 1 s	27 s < 3 s 1 s
Sensitivity[3]		NEO-6G/Q	NEO-6M
	Tracking: Cold starts: Hot starts:	–162 dBm –148 dBm –157 dBm	–161 dBm –147 dBm –156 dBm

[1] All SV @ –130 dBm
[2] Dependent on aiding data connection speed and latency
[3] Demonstrated with a good active antenna

Electrical data

Power supply	2.7 V – 3.6 V (NEO-6Q/6M) 1.75 V – 2.0 V (NEO-6G)
Power consumption	111 mW @ 3.0 V (continuous) 33 mW @ 3.0 V Power Save Mode (1 Hz) 68 mW @ 1.8 V (continuous) 22 mW @ 1.8 V Power Save Mode (1 Hz)
Backup power	1.4 V – 3.6V, 22 µA
Supported antennas	Active and passive

Interfaces

Serial interfaces	1 UART 1 USB V2.0 full speed 12 Mbit/s 1 DDC (I²C compliant) 1 SPI	
Digital I/O	Configurable timepulse 1 EXTINT input for Wakeup	
Serial and I/O	Voltages	2.7 – 3.6 V (NEO-6Q/6M) 1.75 – 2.0 V (NEO-6G)
Timepulse	Configurable	0.25 Hz to 1 kHz
Protocols	NMEA, UBX binary, RTCM	

Specification applies to FW 7

Package

24 pin LCC (Leadless Chip Carrier): 12.2 x 16.0 x 2.4 mm, 1.6 g

Pinout

Environmental data, quality & reliability

Operating temp.	–40° C to 85° C
Storage temp.	–40° C to 85° C

RoHS compliant (lead-free)

Qualification according to ISO 16750

Manufactured in ISO/TS 16949 certified production sites

Support products

u-blox 6 Evaluation Kits:

Easy-to-use kits to get familiar with u-blox 6 positioning technology, evaluate functionality, and visualize GPS performance.

EVK-6H:	u-blox6 Evaluation Kit with TCXO, suitable for NEO-6G, NEO-6Q
EVK-6P:	u-blox6 Evaluation Kit with crystal, suitable for NEO-6M

Ordering information

NEO-6G-0	u-blox 6 GPS Module, 1.8V, TCXO, 12x16mm, 250 pcs/reel
NEO-6M-0	u-blox 6 GPS Module, 12x16mm, 250 pcs/reel
NEO-6Q-0	u-blox 6 GPS Module, TCXO, 12x16mm, 250 pcs/reel

Available as samples and tape on reel (250 pieces)

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